
TRIGONOMETRY

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1. INVERSE TRIG FUNCTIONS

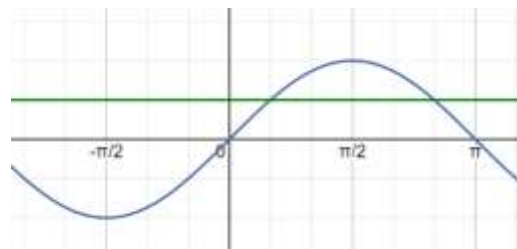
1.1 Definition and Basics

A. Inverse Sine Functions

After you are done with this section you should not only understand the inverse trigonometric functions, but also have their *graphs, domain, range* and *properties* memorized.

1.1: Inverse Sine Function

- Does the function $y = \sin x$ meet the horizontal line test for an invertible function? Hence, does it have an inverse?
- Identify a restricted domain for the sin function such that the entire range of $\sin x$ is achieved, 0 lies in the domain, and the function meets the horizontal line test. What quadrants does this represent on the unit circle?
- Graph $y = \sin^{-1} x$ by reflecting $y = \sin x$ across the line $y = x$. State its domain and range.
- When is $\sin^{-1} x > 0$? < 0 ? neither positive nor negative?



Part A

The sine function does not satisfy the horizontal line test. Hence, there is no inverse for the function:

$$y = \sin x$$

However, we can make the function invertible by restricting the domain.

Part B

Range of $\sin x$ is $[-1, 1]$

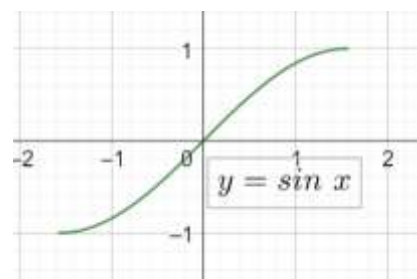
The restricted domain is:

$$\underbrace{y = -1 \Rightarrow x = -\frac{\pi}{2}}_{\text{Start Point}}, \quad \underbrace{y = 1 \Rightarrow x = \frac{\pi}{2}}_{\text{End Point}}$$

It represents

$$0 < x < \frac{\pi}{2} \Leftrightarrow 0^\circ < x < 90^\circ \Leftrightarrow \text{Ist Quadrant}$$

$$-\frac{\pi}{2} < x < 0 \Leftrightarrow -90^\circ < x < 0^\circ \Leftrightarrow \text{IVth Quadrant}$$



Part C

$$y = \sin^{-1} x, \quad \underbrace{-1 \leq x \leq 1}_{\text{Domain}}, \quad \underbrace{-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}}_{\text{Range}}$$

Part D

$\sin^{-1} x$ is positive when x is in the first quadrant in the unit circle. We can write this in interval notation:

$$(0, 90^\circ] \Leftrightarrow \left(0, \frac{\pi}{2}\right]$$

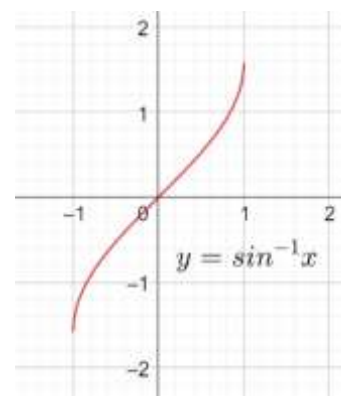
$\sin^{-1} x$ is negative when x is in the fourth quadrant in the unit circle. We can write this in interval notation:

$$[-90^\circ, 0) \Leftrightarrow \left[-\frac{\pi}{2}, 0\right)$$

In other words, for $x \in D_{\sin^{-1} x}$:

$$x > 0 \Rightarrow \sin^{-1} x > 0$$

$$x < 0 \Rightarrow \sin^{-1} x < 0$$



1.2: Inverse Sine Function

The sine inverse function satisfies the cancellation equations:

$$\begin{aligned}\sin(\sin^{-1} x) &= x, & -1 \leq x \leq 1 \\ \sin^{-1}(\sin x) &= x, & -\frac{\pi}{2} \leq x \leq \frac{\pi}{2}\end{aligned}$$

Example 1.3

Evaluating $\sin^{-1}$

Answer in both degrees and radians. If your answer is negative, write your final answer as an equivalent positive angle.

- A. $\sin^{-1}\left(\frac{1}{2}\right)$
- B. $\sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$
- C. $\sin^{-1}\left(\frac{1}{\sqrt{2}}\right)$
- D. $\sin^{-1}\left(-\frac{1}{\sqrt{2}}\right)$
- E. $\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$
- F. $\sin^{-1}\left(-\frac{1}{2}\right)$

Basic Cancellations

Evaluate. Write your answer in the same notation (degrees/radians) as in the question.

- G. $\sin\left(\sin^{-1}\frac{1}{2}\right)$
- H. $\sin\left(\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)$
- I. $\sin(\sin^{-1} 0.24)$
- J. $\sin(\sin^{-1} 1.2)$
- K. $\sin^{-1}(\sin 45^\circ)$
- L. $\sin^{-1}\left(\sin \frac{\pi}{3}\right)$
- M. $\sin^{-1}\left(\sin \frac{11}{6}\pi\right)$

Cancellations using $\sin(180 - \theta) = \sin \theta$

Evaluate. Write your answer in the same notation (degrees/radians) as in the question.

- N. $\sin^{-1}(\sin 120^\circ)$
- O. $\sin^{-1}(\sin 119^\circ)$
- P. $\sin^{-1}\left(\sin \frac{3\pi}{5}\right)$

Evaluating $\sin^{-1}$

$$\begin{aligned}\sin^{-1}\left(\frac{1}{2}\right) &= 30^\circ = \frac{\pi}{6} \\ \sin^{-1}\left(\frac{\sqrt{3}}{2}\right) &= 60^\circ = \frac{\pi}{3} \\ \sin^{-1}\left(\frac{1}{\sqrt{2}}\right) &= 45^\circ = \frac{\pi}{4} \\ \sin^{-1}\left(-\frac{1}{\sqrt{2}}\right) &= -45^\circ = -\frac{\pi}{4} = \frac{3\pi}{4} \\ \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right) &= -60^\circ = -\frac{\pi}{3} = \frac{2\pi}{3} \\ \sin^{-1}\left(-\frac{1}{2}\right) &= -30^\circ = -\frac{\pi}{6} = \frac{5\pi}{6}\end{aligned}$$

First and Fourth Quadrants

$$\sin\left(\sin^{-1}\frac{1}{2}\right) = \frac{1}{2}$$

$$\begin{aligned}\sin\left(\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right) &= -\frac{\sqrt{3}}{2} \\ \sin(\sin^{-1} 0.24) &= 0.24\end{aligned}$$

$$\sin(\sin^{-1} 1.2)$$

→ Not defined since 1.2 is not in the domain of $\sin^{-1} x$

$$\sin^{-1}(\sin 45^\circ) = 45^\circ$$

$$\sin^{-1}\left(\sin \frac{\pi}{3}\right) = \frac{\pi}{3}$$

$$\sin^{-1}\left(\sin \frac{11}{6}\pi\right) = \frac{11}{6}\pi$$

Using $\sin(180 - \theta)$

$$\sin^{-1}(\sin(180 - 60)) = \sin^{-1}(\sin 60) = 60^\circ$$

$$\sin^{-1}(\sin(180 - 61)) = \sin^{-1}(\sin 61) = 61^\circ$$

$$\sin^{-1}\left(\sin\left(\pi - \frac{3\pi}{5}\right)\right) = \sin^{-1}\left(\sin\left(\frac{2\pi}{5}\right)\right) = \frac{2\pi}{5}$$

Example 1.4

Evaluate $h(x)$, $0 \leq x \leq 2\pi$ if:

$$f(x) = \sin x, \quad g(x) = \sin^{-1} x, \quad h(x) = g(f(x))$$

$$h(x) = g(f(x)) = \sin^{-1}(\sin x)$$

If x is in the first or the fourth quadrant, then the function and its inverse will cancel:

$$0 \leq x \leq \frac{\pi}{2}, \frac{3\pi}{4} \leq x \leq 2\pi \Rightarrow h(x) = x$$

If x is in the second or the third quadrant, then x is not in the domain of $\sin^{-1} x$, so we can't cancel directly. We need to find a corresponding value:

$$\sin x = \sin(180 - x) \\ h(x) = \sin^{-1}(\sin x) = \sin^{-1}(\sin 180 - x) = 180 - x$$

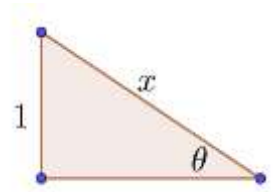
$$h(x) = \begin{cases} x, & 0 \leq x \leq \frac{\pi}{2}, \frac{3\pi}{4} \leq x \leq 2\pi \\ 180 - x, & \frac{\pi}{2} \leq x < \frac{3\pi}{4} \end{cases}$$

1.5: Reciprocal Property

$$\sin^{-1} x = \csc^{-1} \frac{1}{x}, x \in [-1, 1] \\ \csc^{-1} x = \sin^{-1} \frac{1}{x}, x \in \mathbb{R} - (-1, 1)$$

$$\theta = \csc^{-1} x, \quad -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$\csc \theta = x \Rightarrow \sin \theta = \frac{1}{x} \Rightarrow \theta = \sin^{-1} \frac{1}{x}$$



Example 1.6

$$\sin \frac{\pi}{3} = \frac{\sqrt{3}}{2} \Rightarrow \sin^{-1} \frac{\sqrt{3}}{2} = \frac{\pi}{3}$$

$$\csc \frac{\pi}{3} = \frac{2}{\sqrt{3}} \Rightarrow \csc^{-1} \frac{2}{\sqrt{3}} = \sin^{-1} \left(\frac{\sqrt{3}}{2} \right) = \frac{\pi}{3}$$

$$\csc^{-1} \frac{2}{\sqrt{3}} = \frac{1}{\sin^{-1} \frac{2}{\sqrt{3}}} \approx \frac{1}{\sin^{-1} 1.73} \rightarrow \text{Not Defined}$$

1.7: Odd Function

An odd function is a function that satisfies $f(-x) = -f(x)$. The inverse sine function is an odd function. That is, it satisfies:

$$\sin^{-1}(-y) = -\sin^{-1} y$$

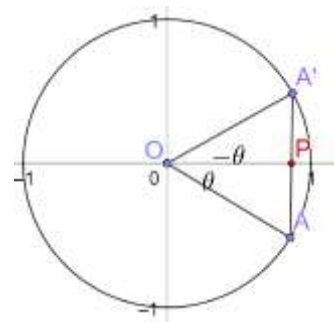
$$\theta = \sin^{-1}(-y)$$

Apply the sin function both sides:

$$\sin \theta = -y \Rightarrow y = -\sin \theta \Rightarrow y = \sin(-\theta)$$

Apply the $\sin^{-1}$ function to both sides:

$$\sin^{-1} y = -\theta \Rightarrow \theta = -\sin^{-1} y \Rightarrow \sin^{-1}(-y) = -\sin^{-1} y$$

**B. Inverse Cosine Function****1.8: Inverse Cosine Function****Example 1.9****Evaluating $\cos^{-1}$**

Answer in both degrees and radians. If your answer is negative, write your final answer as an equivalent positive angle.

- A. $\cos^{-1}\left(\frac{1}{2}\right)$
- B. $\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)$
- C. $\cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$
- D. $\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right)$
- E. $\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)$
- F. $\cos^{-1}\left(-\frac{1}{2}\right)$

Basic Cancellations

Evaluate. Write your answer in the same notation (degrees/radians) as in the question.

- G. $\cos\left(\cos^{-1}\frac{1}{2}\right)$
- H. $\cos\left(\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)$
- I. $\cos(\cos^{-1} 0.24)$
- J. $\cos(\cos^{-1} 1.2)$
- K. $\cos^{-1}(\cos 45^\circ)$
- L. $\cos^{-1}\left(\cos\frac{\pi}{3}\right)$
- M. $\cos^{-1}\left(\cos\frac{11}{6}\pi\right)$

Cancellations using $\cos(180 - \theta) = -\cos \theta$

Evaluate. Write your answer in the same notation (degrees/radians) as in the question.

- N. $\cos^{-1}(\cos 120^\circ)$
- O. $\cos^{-1}(\cos 119^\circ)$
- P. $\cos^{-1}\left(\cos\frac{3\pi}{5}\right)$

$$\begin{aligned}\cos^{-1}\left(\frac{1}{2}\right) &= 60^\circ \\ \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) &= 30^\circ \\ \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) &= 45^\circ\end{aligned}$$

$$\begin{aligned}\cos^{-1}\left(-\frac{1}{\sqrt{2}}\right) &= 135^\circ \\ \cos^{-1}\left(-\frac{\sqrt{3}}{2}\right) &= 150^\circ\end{aligned}$$

C. Inverse Tan Function**1.10: Inverse Tan Function**

Example 1.11**Evaluating $\cos^{-1}$**

Answer in both degrees and radians. If your answer is negative, write your final answer as an equivalent positive angle.

- A. $\tan^{-1}(\sqrt{3})$
- B. $\tan^{-1}\left(\frac{\sqrt{3}}{3}\right)$
- C. $\tan^{-1}(1)$
- D. $\tan^{-1}(-\sqrt{3})$
- E. $\tan^{-1}\left(-\frac{\sqrt{3}}{3}\right)$
- F. $\tan^{-1}(-1)$

Basic Cancellations

Evaluate. Write your answer in the same notation (degrees/radians) as in the question.

- G. $\sin\left(\sin^{-1}\frac{1}{2}\right)$
- H. $\sin\left(\sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)\right)$
- I. $\sin(\sin^{-1} 0.24)$
- J. $\sin(\sin^{-1} 1.2)$
- K. $\sin^{-1}(\sin 45^\circ)$
- L. $\sin^{-1}\left(\sin\frac{\pi}{3}\right)$
- M. $\sin^{-1}\left(\sin\frac{11}{6}\pi\right)$

Cancellations using $\sin(180 - \theta) = \sin \theta$

Evaluate. Write your answer in the same notation (degrees/radians) as in the question.

- N. $\sin^{-1}(\sin 120^\circ)$
- O. $\sin^{-1}(\sin 119^\circ)$
- P. $\sin^{-1}\left(\sin\frac{3\pi}{5}\right)$

1.12: Inverse Function-Inverse Cofunction Identities

$$\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$$

$$\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x$$

$$\csc^{-1} x = \frac{\pi}{2} - \sec^{-1} x$$

$$\begin{aligned} \theta &= \sin^{-1}(x) \Rightarrow \sin \theta = x \\ \cos\left(\frac{\pi}{2} - \theta\right) &= x \Rightarrow \frac{\pi}{2} - \theta = \cos^{-1} x \Rightarrow \frac{\pi}{2} = \theta + \cos^{-1} x = \sin^{-1}(x) + \cos^{-1} x \end{aligned}$$

D. Basic Applications**Example 1.13**

$$\tan^{-1}\left(\frac{1+\sqrt{3}}{3+\sqrt{3}}\right) + \sec^{-1}\left(\sqrt{\frac{8+4\sqrt{3}}{6+3\sqrt{3}}}\right)$$

is equal to: (JEE-M 2023)

$$\begin{aligned} \tan^{-1}\left(\frac{1+\sqrt{3}}{\sqrt{3}(\sqrt{3}+1)}\right) &= \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = \frac{\pi}{6} \\ \sec^{-1}\left(\frac{2\sqrt{2+\sqrt{3}}}{\sqrt{3}\sqrt{2+\sqrt{3}}}\right) &= \sec^{-1}\left(\frac{2}{\sqrt{3}}\right) = \cos^{-1}\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{6} \end{aligned}$$

Hence, the final answer is:

$$\frac{\pi}{6} + \frac{\pi}{6} = \frac{\pi}{3}$$

E. Reference Triangles

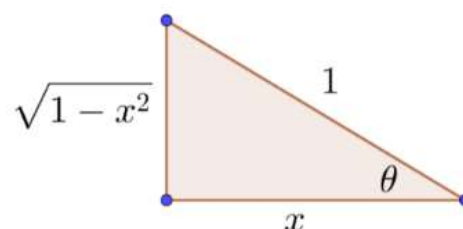
In many places, it is useful to convert from one trigonometric function to another, and reference triangles let you do exactly that. Here, we use reference triangles to eliminate inverse trigonometric functions from an expression.

Example 1.14

$$\sin(\cos^{-1} x)$$

Use a change of variable. Let:

$$\begin{aligned}\theta &= \cos^{-1} x \\ \cos \theta &= x = \frac{x}{1} = \frac{Adj}{Hyp}\end{aligned}$$



Draw a right-angled triangle with θ as one of the angles, and find the opposite side:

$$Opp = \sqrt{Hyp^2 - Adj^2} = \sqrt{1 - x^2}$$

Finally:

$$\begin{aligned}\sin(\cos^{-1} x) &= \sin \theta = \frac{\sqrt{1-x^2}}{1} = \sqrt{1-x^2} \\ 1-x^2 &\geq 0 \Rightarrow 1 \geq x^2 \Rightarrow x^2 \leq 1 \Rightarrow x \in [-1, 1]\end{aligned}$$

Example 1.15

$$\cos(\sin^{-1} x)$$

Let:

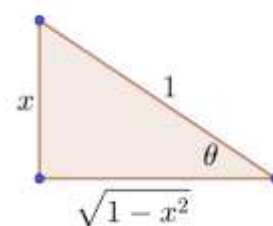
$$\theta = \sin^{-1} x$$

Take sin of both sides:

$$\sin \theta = x = \frac{x}{1}$$

Draw a reference triangle. Then:

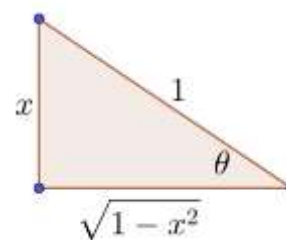
$$\cos(\sin^{-1} x) = \cos \theta = \sqrt{1-x^2}$$



Example 1.16

$$\tan(\sin^{-1} x)$$

$$\begin{aligned}\tan(\sin^{-1} x) &= \frac{\sin(\sin^{-1} x)}{\cos(\sin^{-1} x)} = \frac{x}{\sqrt{1-x^2}} \\ \theta &= \sin^{-1} x \Rightarrow \sin \theta = x = \frac{x}{1} \\ \tan(\sin^{-1} x) &= \tan \theta = \frac{x}{\sqrt{1-x^2}}\end{aligned}$$



Example 1.17

$$\sec(\cos^{-1} x)$$

Let

$$\theta = \cos^{-1} x \Rightarrow \cos \theta = x \Rightarrow \sec \theta = \frac{1}{\cos \theta} = \frac{1}{x}$$

$$\sec(\cos^{-1} x) = \sec \theta = \frac{1}{x}$$

Example 1.18

$$\tan\left(\sin^{-1} \frac{x}{x+1}\right)$$

$$\theta = \sin^{-1} \frac{x}{x+1} \Rightarrow \sin \theta = \frac{x}{x+1}$$

$$\tan\left(\sin^{-1} \frac{x}{x+1}\right) = \tan \theta = \frac{x}{\sqrt{2x+1}}$$

Example 1.19

$$\tan\left(\sec^{-1} \frac{x}{2}\right)$$

$$\theta = \sec^{-1} \frac{x}{2} \Rightarrow \sec \theta = \frac{x}{2} \Rightarrow \cos \theta = \frac{2}{x}$$

$$\begin{aligned} \text{opp side} &= \sqrt{x^2 - 4} \\ \tan \theta &= \frac{\sqrt{x^2 - 4}}{2} \end{aligned}$$

Example 1.20

Show that if $\sin(\sin^{-1} x + \cos^{-1} y) = 2xy$, then x and y both lie on the unit circle.

$$LHS = xy + \sqrt{(1-x^2)(1-y^2)}$$

$$\begin{aligned} xy + \sqrt{(1-x^2)(1-y^2)} &= 2xy \\ \sqrt{(1-x^2)(1-y^2)} &= xy \\ (1-x^2)(1-y^2) &= x^2y^2 \\ 1-x^2-y^2+x^2y^2 &= x^2y^2 \\ 1-x^2-y^2 &= 0 \\ 1 &= x^2+y^2 \end{aligned}$$

Example 1.21: Identities

We can also simplify trigonometric functions of inverse trigonometric functions using trigonometric identities.

- A. $\sin(\cos^{-1} x)$
- B. $\cos(\sin^{-1} x)$

Part A

Substitute $\theta = \cos^{-1} x$ in the Pythagorean Identity $\sin^2 \theta + \cos^2 \theta = 1$:

$$\begin{aligned} \sin^2(\cos^{-1} x) + \cos^2(\cos^{-1} x) &= 1 \\ \sin^2(\cos^{-1} x) + x^2 &= 1 \\ \sin^2(\cos^{-1} x) &= 1 - x^2 \\ \sin(\cos^{-1} x) &= \sqrt{1 - x^2} \end{aligned}$$

Part B

Let:

$$\cos^2 \theta = 1 - \sin^2 \theta = 1 - x^2 \Rightarrow \cos \theta = \sqrt{1 - x^2}$$

$$\theta = \sin^{-1} x$$

$$\cos(\sin^{-1} x) = \cos \theta = \sqrt{1 - x^2}$$

1.2 Properties

A. Basics

Example 1.22: Addition Formula

$$\cos\left(\tan^{-1} \frac{4}{3} - \cos^{-1} \frac{5}{13}\right)$$

Use the formula $\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$:

$$= \cos\left(\tan^{-1} \frac{4}{3}\right) \cos\left(\cos^{-1} \frac{5}{13}\right) + \sin\left(\tan^{-1} \frac{4}{3}\right) \sin\left(\cos^{-1} \frac{5}{13}\right)$$

$$\tan \theta = \frac{4}{3} \Rightarrow \theta = \tan^{-1} \frac{4}{3} = \cos^{-1} \frac{3}{5} = \sin^{-1} \frac{4}{5}$$

$$\cos \theta = \frac{5}{13} \Rightarrow \theta = \cos^{-1} \frac{5}{13} = \sin^{-1} \frac{12}{13}$$

$$\cos\left(\tan^{-1} \frac{4}{3}\right) = \cos\left(\cos^{-1} \frac{3}{5}\right) = \frac{3}{5}$$

$$\cos\left(\cos^{-1} \frac{5}{13}\right) = \frac{5}{13}$$

$$\sin\left(\tan^{-1} \frac{4}{3}\right) = \sin\left(\sin^{-1} \frac{4}{5}\right) = \frac{4}{5}$$

$$\sin\left(\cos^{-1} \frac{5}{13}\right) = \sin\left(\sin^{-1} \frac{12}{13}\right) = \frac{12}{13}$$

$$= \frac{3}{5} \times \frac{5}{13} + \frac{4}{5} \times \frac{12}{13} = \frac{3}{13} + \frac{48}{65} = \frac{15}{65} + \frac{48}{65} = \frac{63}{65}$$

Example 1.23: Double Angle Formula

$$\sin(2 \sin^{-1} x)$$

Apply the formula $\sin 2\theta = 2 \sin \theta \cos \theta$:

$$2 \sin(\sin^{-1} x) \cos(\sin^{-1} x)$$

1.24: Tan Addition Identity for Inverses

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x + y}{1 - xy} \right), \quad xy < 1$$

$$\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x - y}{1 + xy} \right), \quad xy > -1$$

Take the tan inverse of both sides of the identity $\tan(\alpha + \beta) = \frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta}$:

$$\alpha + \beta = \tan^{-1} \left(\frac{\tan \alpha + \tan \beta}{1 - \tan \alpha \tan \beta} \right)$$

Let $\tan \alpha = x \Rightarrow \alpha = \tan^{-1} x$, $\tan \beta = y \Rightarrow \beta = \tan^{-1} y$:

$$\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x + y}{1 - xy} \right)$$

The proof of the second identity is similar, but it makes use of

$$\tan(\alpha - \beta) = \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta}$$

Example 1.25

- The value of $\tan \left[\cos^{-1} \frac{4}{5} + \tan^{-1} \frac{2}{3} \right]$ is: (JEE 1983)
- The value of $\sin \left[\cos^{-1} \frac{4}{5} + \tan^{-1} \frac{2}{3} \right]$ is: (CBSE 2019)
- Prove that $\tan^{-1} \frac{1}{5} + \tan^{-1} \frac{1}{7} + \tan^{-1} \frac{1}{3} + \tan^{-1} \frac{1}{8} = \frac{\pi}{4}$ (CBSE 2016)
- If $\tan^{-1} \left(\frac{1}{1+1 \cdot 2} \right) + \tan^{-1} \left(\frac{1}{1+2 \cdot 3} \right) + \dots + \tan^{-1} \left(\frac{1}{1+n(n+1)} \right) = \tan^{-1} \theta$, then find the value of θ . (CBSE 2015)

Part D

Use the identity $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$ to expand each term:

$$\tan^{-1} \left(\frac{2-1}{1+1 \cdot 2} \right) = \tan^{-1} 2 - \tan^{-1} 1$$

$$\tan^{-1} \left(\frac{3-2}{1+2 \cdot 3} \right) = \tan^{-1} 3 - \tan^{-1} 2$$

⋮

$$\tan^{-1} \left(\frac{(n+1)-n}{1+n(n+1)} \right) = \tan^{-1}(n+1) - \tan^{-1} n$$

Note that we have a telescoping series. Add the terms:

$$\tan^{-1}(n+1) - \tan^{-1} 1 = \tan^{-1} \left(\frac{(n+1)-1}{1+1(n+1)} \right) = \tan^{-1} \left(\frac{n}{n+2} \right)$$

B. Applications

1.26:

$$a \sin \theta + b \cos \theta = \sqrt{a^2 + b^2} \sin \left(\theta + \tan^{-1} \left(\frac{b}{a} \right) \right)$$

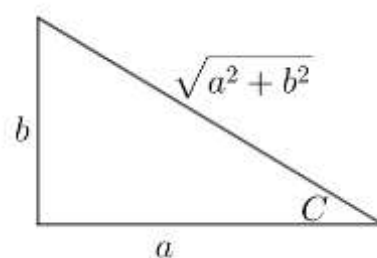
Let

$$C = \tan^{-1} \left(\frac{b}{a} \right) \Rightarrow \tan C = \frac{b}{a}$$

Use the sum formula on the RHS:

$$RHS = \sqrt{a^2 + b^2} [\sin \theta \cos C + \cos \theta \sin C]$$

Substitute $\cos C = \frac{a}{\sqrt{a^2+b^2}}$, $\sin C = \frac{b}{\sqrt{a^2+b^2}}$



$$= \sqrt{a^2 + b^2} \left[\sin \theta \frac{a}{\sqrt{a^2 + b^2}} + \cos \theta \frac{b}{\sqrt{a^2 + b^2}} \right]$$

$$= a \sin \theta + b \cos \theta = LHS$$

Example 1.27

$$\sin \theta + \cos \theta = \sqrt{2} \sin \left(\theta + \frac{\pi}{4} \right)$$

- A. Using the formula
- B. Without the formula

Part A

Using $a = b = 1$

$$\sqrt{a^2 + b^2} = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\tan^{-1} \left(\frac{b}{a} \right) = \tan^{-1} \left(\frac{1}{1} \right) = \frac{\pi}{4}$$

$$\sqrt{a^2 + b^2} \sin \left(\theta + \tan^{-1} \left(\frac{b}{a} \right) \right) = \sqrt{2} \sin \left(\theta + \frac{\pi}{4} \right)$$

Part B

Factor $\sqrt{2}$ in the LHS:

$$LHS = \sqrt{2} \left[(\sin \theta) \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \cos \theta \right]$$

Substitute $\sin \left(\frac{\pi}{4} \right) = \cos \left(\frac{\pi}{4} \right) = \frac{1}{\sqrt{2}}$:

$$= \sqrt{2} \left[(\sin \theta) \cos \left(\frac{\pi}{4} \right) + \sin \left(\frac{\pi}{4} \right) \cos \theta \right]$$

Use the sum formula $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \sin \beta \cos \alpha$

$$= \sqrt{2} \left[\sin \left(\theta + \frac{\pi}{4} \right) \right] = RHS$$

Example 1.28

$$\sqrt{12} \sin 3\alpha + 2 \cos 3\alpha$$

Use a change of variable. Let $3\alpha = \theta$:

$$\sqrt{12} \sin \theta + 2 \cos \theta$$

Using $a = \sqrt{12}, b = 2$

$$\sqrt{a^2 + b^2} = \sqrt{(\sqrt{12})^2 + 2^2} = \sqrt{12 + 4} = \sqrt{16} = 4$$

$$\tan^{-1} \left(\frac{2}{\sqrt{12}} \right) = \tan^{-1} \left(\frac{2}{2\sqrt{3}} \right) = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = \frac{\pi}{6}$$

$$\sqrt{a^2 + b^2} \sin \left(\theta + \tan^{-1} \left(\frac{b}{a} \right) \right) = 4 \sin \left(\theta + \frac{\pi}{6} \right) = 4 \sin \left(3\alpha + \frac{\pi}{6} \right)$$

1.29:

$$a \sin \theta + b \cos \theta = \sqrt{a^2 + b^2} \cos \left(\theta - \tan^{-1} \left(\frac{a}{b} \right) \right)$$

Let

$$C = \tan^{-1}\left(\frac{a}{b}\right) \Rightarrow \tan C = \frac{a}{b}$$

Use the sum formula on the RHS:

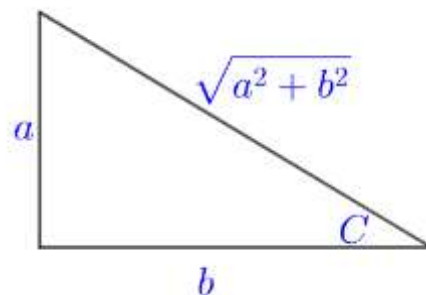
$$RHS = \sqrt{a^2 + b^2} [\cos \theta \cos C + \sin \theta \sin C]$$

Substitute $\cos C = \frac{b}{\sqrt{a^2 + b^2}}$, $\sin C = \frac{a}{\sqrt{a^2 + b^2}}$

$$= \sqrt{a^2 + b^2} \left[\cos \theta \frac{b}{\sqrt{a^2 + b^2}} + \sin \theta \frac{a}{\sqrt{a^2 + b^2}} \right]$$

Simplify:

$$= a \sin \theta + b \cos \theta = LHS$$



Example 1.30

$$\sin \theta + \cos \theta = \sqrt{2} \sin \left(\theta + \frac{\pi}{4} \right)$$

$$\sqrt{a^2 + b^2} \cos \left(\theta - \tan^{-1} \left(\frac{a}{b} \right) \right)$$

Substitute $a = b = 1$

$$= \sqrt{1^2 + 1^2} \cos \left(\theta - \tan^{-1} \left(\frac{1}{1} \right) \right) = \sqrt{2} \cos \left(\theta - \frac{\pi}{4} \right)$$

Example 1.31

$$\sqrt{12} \sin 3\alpha + 2 \cos 3\alpha$$

Using $a = \sqrt{12}, b = 2$:

$$\sqrt{a^2 + b^2} = \sqrt{(\sqrt{12})^2 + 2^2} = \sqrt{12 + 4} = \sqrt{16} = 4$$

$$\tan^{-1} \left(\frac{\sqrt{12}}{2} \right) = \tan^{-1} \left(\frac{2\sqrt{3}}{2} \right) = \tan^{-1}(\sqrt{3}) = \frac{\pi}{3}$$

$$\sqrt{a^2 + b^2} \cos \left(\theta - \tan^{-1} \left(\frac{a}{b} \right) \right) = 4 \cos \left(3\alpha - \frac{\pi}{3} \right)$$

C. Defined Operations

Example 1.32

Let $x \star y = x^2 + y^3$ and $(x \star 1) \star 1 = x \star (1 \star 1)$, then the principal value of $2 \sin^{-1} \left(\frac{x^4 + x^2 - 2}{x^4 + x^2 + 2} \right)$ is: (JEE-M 2022)

$$\begin{aligned} (x^2 + 1) \star 1 &= x \star 2 \\ (x^2 + 1)^2 + 1 &= x^2 + 8 \\ x^4 + 2x^2 + 2 &= x^2 + 8 \end{aligned}$$

Substitute $x^4 + x^2 = 6$:

$$2 \sin^{-1} \left(\frac{x^4 + x^2 - 2}{x^4 + x^2 + 2} \right) = 2 \sin^{-1} \left(\frac{6 - 2}{6 + 2} \right) = 2 \sin^{-1} \left(\frac{1}{2} \right) = 2 \cdot \frac{\pi}{6} = \frac{\pi}{3}$$

1.3 Equations

A. Basics

Example 1.33

The number of solutions of $\sin^{-1} x = 2 \tan^{-1} x$ is equal to: (JEE-M 2023)

Let

$$\theta = 2 \tan^{-1} x \Rightarrow \frac{\theta}{2} = \tan^{-1} x$$

Take the *tan* of the both sides and draw a reference triangle:

$$\tan \frac{\theta}{2} = x \Rightarrow \sin \frac{\theta}{2} = \frac{x}{\sqrt{1+x^2}}, \quad \cos \frac{\theta}{2} = \frac{1}{\sqrt{1+x^2}}$$

$$\sin \theta = 2 \sin \frac{\theta}{2} \cos \frac{\theta}{2} = 2 \cdot \frac{x}{\sqrt{1+x^2}} \cdot \frac{1}{\sqrt{1+x^2}} = \frac{2x}{1+x^2}$$

Substitute $\theta = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$ in the given equation:

$$\sin^{-1} x = \sin^{-1} \left(\frac{2x}{1+x^2} \right)$$

$$x = \frac{2x}{1+x^2}$$

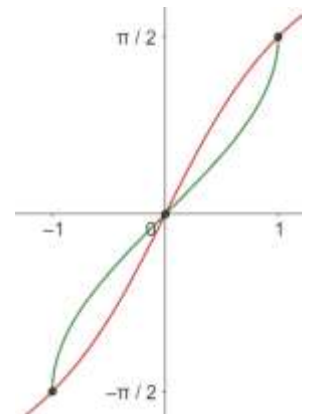
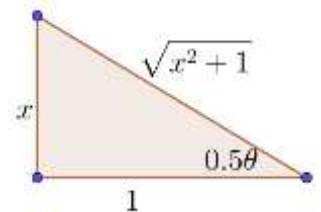
$$x + x^3 = 2x$$

$$x^3 - x = 0$$

$$x(x^2 - 1) = 0$$

$$x \in \{-1, 0, 1\}$$

3 Solutions

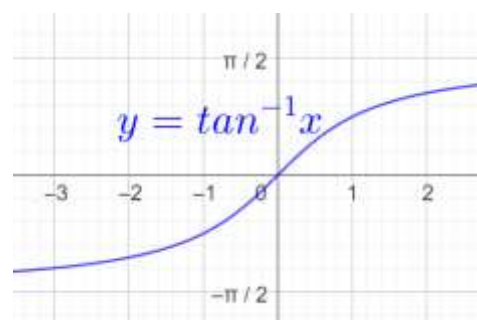
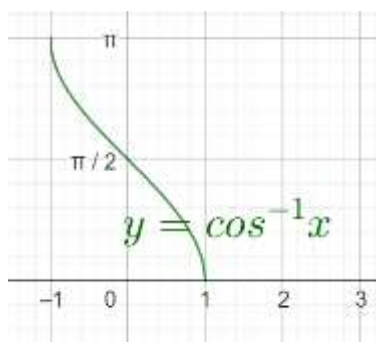
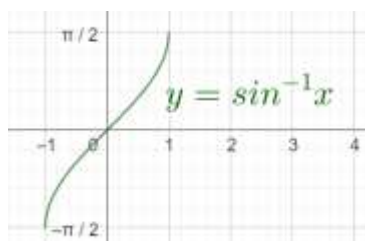


1.4 Domain and Range

A. Multiple Domains

Domain and Range

Function	Domain	Range
$y = \sin^{-1} x$	$-1 \leq x \leq 1$	$-\frac{\pi}{2} \leq y \leq \frac{\pi}{2}$
$y = \cos^{-1} x$	$-1 \leq x \leq 1$	$0 \leq y \leq \pi$
$y = \tan^{-1} x$	$-\infty < x < \infty$	$-\frac{\pi}{2} < y < \frac{\pi}{2}$



1.34: Intersection of Multiple Domains

If there is more than one part to the function, each part must be checked separately.

The final domain is the *intersection* of all the parts.

Example 1.35

The domain of the function below is: (JEE-M 2002, 2004)

$$f(x) = \frac{\sin^{-1}(x-3)}{\sqrt{9-x^2}}$$

Expression inside the square root must be positive. (Zero not allowed since it is in the denominator).

$$9 - x^2 > 0$$

$$x^2 < 9$$

$$-3 < x < 3$$

The input for $\sin^{-1} x$ must be between -1 and 1

$$-1 \leq x - 3 \leq 1$$

$$2 \leq x \leq 4$$

Take the intersection of the above two:

$$[2, 3)$$

Example 1.36

The interval for which the function below is defined is: (JEE-M 2007, Adapted)

$$f(x) = 4^{-x^2} + \cos^{-1}\left(\frac{x}{3} - 1\right) + \log(\cos x)$$

4^{-x^2} has no restrictions.

$\cos^{-1} x$ must be between -1 and 1 .

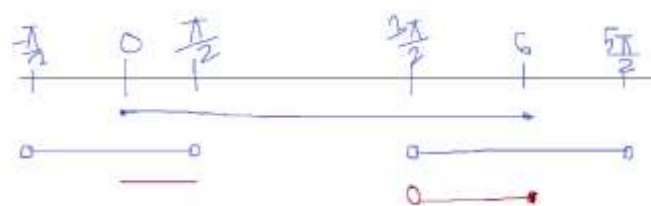
$$-1 \leq \frac{x}{3} - 1 \leq 1 \Rightarrow 0 \leq x \leq 6$$

The input for the log function must be positive:

$$\cos x > 0 \Rightarrow x \in QI, QIV \Rightarrow \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, \frac{5\pi}{2}\right)$$

The final answer is:

$$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{3\pi}{2}, 6\right]$$



B. Compound Inequalities

1.37: Compound Inequality

The compound inequality $a < b < c$ can be split into the equivalent equalities

$$a < b, \quad b < c$$

Example 1.38

The domain of the function

$$f(x) = \sin^{-1} \left(\frac{x^2 - 3x + 2}{x^2 + 2x + 7} \right) \quad (\text{JEE Main, July 29, 2022 - II})$$

$$-1 \leq \frac{x^2 - 3x + 2}{x^2 + 2x + 7} \leq 1$$

Split into two inequalities:

$$\begin{aligned} \frac{x^2 - 3x + 2}{x^2 + 2x + 7} &\leq 1 \\ x^2 - 3x + 2 &\leq x^2 + 2x + 7 \\ -5x &\leq 5 \\ x &\geq -1 \end{aligned}$$

$$\begin{aligned} \frac{x^2 - 3x + 2}{x^2 + 2x + 7} &\geq -1 \\ x^2 - 3x + 2 &\geq -x^2 - 2x - 7 \\ 2x^2 - x + 9 &\geq 0 \end{aligned}$$

$$b^2 - 4ac = (-1)^2 - 4(2)(9) < 0 \Rightarrow \text{All real numbers are valid}$$

Hence, the final answer is the intersection of the above two:

$$x \in [-1, \infty)$$

Example 1.39

The domain of

$$f(x) = \cos^{-1} \left(\frac{x^2 - 4x + 2}{x^2 + 3} \right) \quad (\text{JEE Main, July 28, 2022 - I})$$

The condition for the domain of $\arccos$ is

$$-1 \leq \frac{x^2 - 4x + 2}{x^2 + 3} \leq 1$$

Since $x^2 + 3 > 0$, multiply throughout by $x^2 + 3$:

$$\begin{aligned} -x^2 - 3 &\leq x^2 - 4x + 2 \leq x^2 + 3 \\ -x^2 - 5 &\leq x^2 - 4x \leq x^2 + 1 \end{aligned}$$

Split the inequality. First part is:

$$\begin{aligned} x^2 - 4x &\leq x^2 + 1 \\ -4x &\leq 1 \\ x &\geq -\frac{1}{4} \end{aligned}$$

Second part is:

$$\begin{aligned} -x^2 - 5 &\leq x^2 - 4x \\ 2x^2 - 4x + 5 &\geq 0 \\ \text{Discriminant} &= (-4)^2 - 4(2)(5) = 16 - 40 < 0 \\ &\text{All Real Numbers} \end{aligned}$$

The final answer is:

$$\begin{aligned} x &\geq -\frac{1}{4} \\ &\left[-\frac{1}{4}, \infty\right) \end{aligned}$$

C. Multiple Domains with Compound Inequalities

Example 1.40

15: The domain of the function below is: (JEE-M 2022)

$$f(x) = \frac{\cos^{-1}\left(\frac{x^2 - 5x + 6}{x^2 - 9}\right)}{\ln(x^2 - 3x + 2)}$$

Example 1.41

If the domain of the function

$$f(x) = \ln(4x^2 + 11x + 6) + \sin^{-1}(4x + 3) + \cos^{-1}\left(\frac{10x + 6}{3}\right)$$

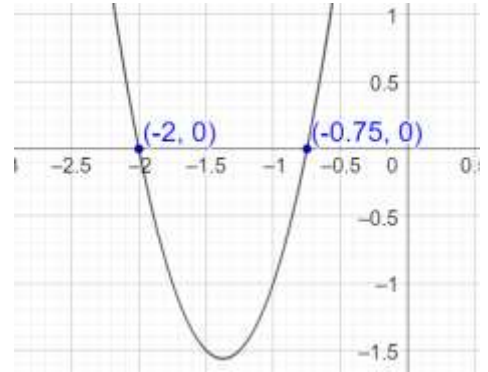
Is $(\alpha, \beta]$, then $36|\alpha + \beta| =$ (JEE-M 2023)

The input for a logarithm must be positive.

$$4x^2 + 11x + 6 > 0$$

$$(4x + 3)(x + 2) = 0 \Rightarrow x \in \left\{-2, -\frac{3}{4}\right\}$$

$$(-\infty, -2) \cup \left(-\frac{3}{4}, \infty\right)$$



The input for $\sin^{-1} x$ must be from -1 to 1 :

$$-1 \leq 4x + 3 \leq 1$$

$$-4 \leq 4x \leq -2$$

$$-1 \leq x \leq -\frac{1}{2}$$

$$x \in \left[-1, -\frac{1}{2}\right]$$

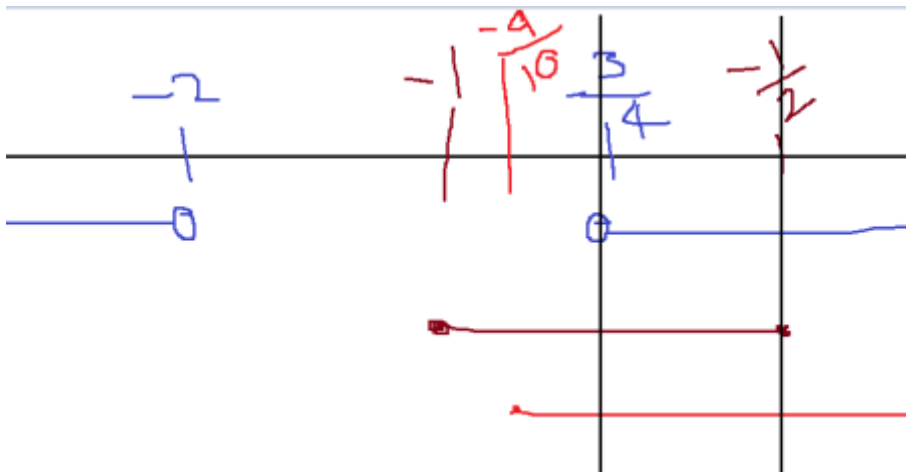
The input for $\cos^{-1} x$ must be from -1 to 1 :

$$\frac{10x + 6}{3} \in [-1, 1]$$

$$10x + 6 \in [-3, 3]$$

$$10x \in [-9, -3]$$

$$x \in \left[-\frac{9}{10}, -\frac{3}{10}\right]$$



The intersection of the three individual domains is:

$$\left(-\frac{3}{4}, -\frac{1}{2}\right]$$

$$36|\alpha + \beta| = 36\left|-\frac{3}{4} - \frac{1}{2}\right| = 36 \cdot \frac{5}{4} = 45$$

Example 1.42

The domain of the function below is: (JEE-M 2021)

$$f(x) = \sin^{-1}\left(\frac{3x^2 + x - 1}{(x - 1)^2}\right) + \cos^{-1}\left(\frac{x - 1}{x + 1}\right)$$

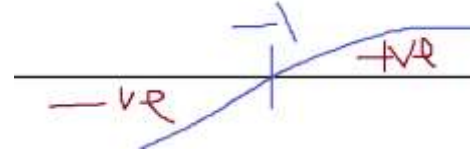
$\cos^{-1} x$ must be between -1 and 1

Split the inequality $-1 \leq \frac{x-1}{x+1} \leq 1$ into two parts.

First Part

$$\frac{x-1}{x+1} \leq 1 \Rightarrow \frac{-2}{x+1} \leq 0 \Rightarrow \frac{2}{x+1} \geq 0$$

Degree = $0 - 1 \rightarrow \text{Odd} \Rightarrow -ve \text{ at } -\infty$
 Critical Points $\in \{-1\}$



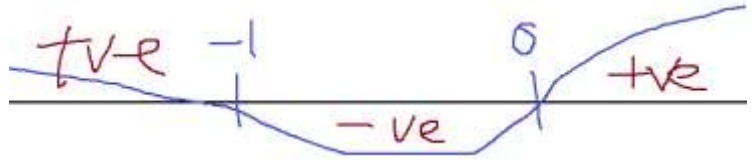
Since the expression is not defined at $x = -1$, remove this from the solution set:

$$(-1, \infty)$$

Second Part

$$\frac{x-1}{x+1} \geq -1 \Rightarrow \frac{2x}{x+1} \geq 0$$

Critical Points $\in \{0, -1\}$
 Degree = $1 - 1 = 0 \Rightarrow \text{Even} \Rightarrow +ve \text{ at } -\infty$

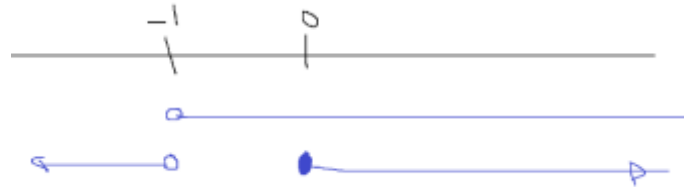


Since the expression is not defined at $x = -1$, remove this from the solution set:

$$(-\infty, -1) \cup [0, \infty)$$

Take the intersection of the solutions from the first and second part:

$$[0, \infty)$$



$\sin^{-1} x$ must be between -1 and 1

$$-1 \leq \frac{3x^2 + x - 1}{(x-1)^2} \leq 1$$

$x = 1$ is a critical point since it makes the denominator zero.

Since $(x-1)^2 > 0$, multiply throughout by $(x-1)^2 = (x^2 - 2x + 1)$:

$$-(x^2 - 2x + 1) \leq 3x^2 + x - 1 \leq (x^2 - 2x + 1)$$

Split the inequality into two parts.

First Part

$$3x^2 + x - 1 \leq (x^2 - 2x + 1)$$

$$2x^2 + 3x - 2 \leq 0$$

$$(2x - 1)(x + 2) \leq 0$$

Critical points $\in$ $\frac{-2}{\text{Odd Multiplicity}}$, $\frac{1}{2}$, $\frac{1}{\text{Even Multiplicity}}$

Degree = $2 - 2 = 0 \Rightarrow \text{Even}$
 $\left[-2, \frac{1}{2}\right]$



Second Part

$$3x^2 + x - 1 \geq -(x^2 - 2x + 1)$$

$$x(4x - 1) \geq 0$$

Critical points $\in$ $\frac{0}{\text{Odd Multiplicity}}$, $\frac{1}{4}$, $\frac{1}{\text{Even Multiplicity}}$



$$\text{Degree} = 2 - 2 = 0 \Rightarrow \text{Even} \Rightarrow +ve \text{ at } -\infty$$

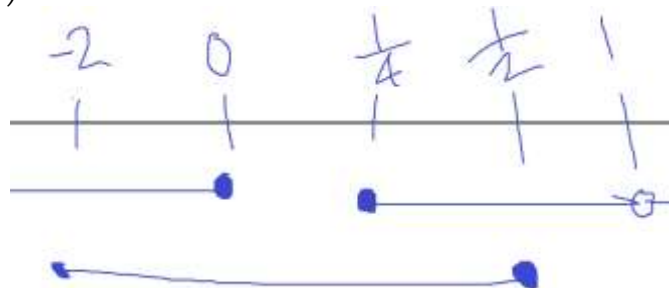
$$(-\infty, 0] \cup \left[\frac{1}{4}, 1\right) \cup (1, \infty)$$

Take the intersection of the first and second part:

$$[-2, 0] \cup \left[\frac{1}{4}, \frac{1}{2}\right]$$

Take the intersection of the above with the solution set from the $\cos^{-1} x$ part $[0, \infty)$

$$\{0\} + \left[\frac{1}{4}, \frac{1}{2}\right]$$



D. Square Roots

Example 1.43

Determine the domain of the function: (JEE-M 2021)

$$f(x) = \frac{\cos^{-1} \sqrt{x^2 - x + 1}}{\sqrt{\sin^{-1} \left(\frac{2x - 1}{2} \right)}}$$

E. Nested Functions

Example 1.44

The domain of the function: (JEE-M 2022)

$$\cos^{-1} \left(\frac{2 \sin^{-1} \left(\frac{1}{4x^2 - 1} \right)}{\pi} \right)$$

Expression in the fraction $\frac{1}{4x^2 - 1}$ cannot have a zero denominator:

$$4x^2 - 1 = 0 \Rightarrow x^2 = \frac{1}{4} \Rightarrow x \in \left\{ -\frac{1}{2}, \frac{1}{2} \right\}$$

Outside Function:

Check the domain of the outside function:

$$-1 \leq \frac{2 \sin^{-1} \left(\frac{1}{4x^2 - 1} \right)}{\pi} \leq 1$$

Multiply both sides by $\frac{\pi}{2}$:

$$-\frac{\pi}{2} \leq \sin^{-1} \left(\frac{1}{4x^2 - 1} \right) \leq \frac{\pi}{2}$$

Inside Function:

Take the $\sin$ of all parts. Note this is precisely the domain requirement of the inside function:

$$-1 \leq \frac{1}{4x^2 - 1} \leq 1$$

Split the inequality.

Case I: Subtract 1 from both sides of $\frac{1}{4x^2-1} \leq 1$

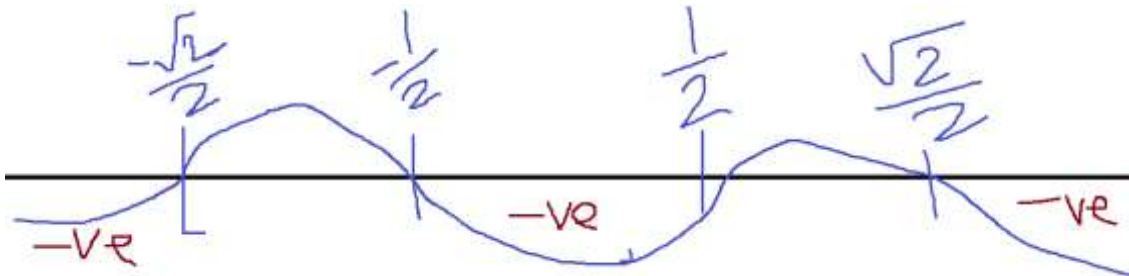
$$\begin{aligned}\frac{1}{4x^2-1} - 1 &\leq 0 \\ \frac{1}{4x^2-1} + \frac{-4x^2+1}{4x^2-1} &\leq 0 \\ \frac{-4x^2+2}{4x^2-1} &\leq 0\end{aligned}$$

Divide by 4 both sides:

$$\frac{-x^2 + \frac{1}{2}}{\left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right)} \leq 0 \Rightarrow \text{Roots} \in \left\{-\frac{\sqrt{2}}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{\sqrt{2}}{2}\right\}$$

Check with -2 :

$$\frac{-x^2 + \frac{1}{2}}{\left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right)} = \frac{-(-2)^2 + \frac{1}{2}}{\left(-2 + \frac{1}{2}\right)\left(-2 - \frac{1}{2}\right)} = \frac{-ve}{(-ve)(-ve)} = -ve$$



The solution set for the interval is:

$$\text{Case I: } \left(-\infty, -\frac{\sqrt{2}}{2}\right] \cup \left[-\frac{1}{2}, \frac{1}{2}\right] \cup \left[\frac{\sqrt{2}}{2}, \infty\right)$$

Case II Add 1 to both sides of $\frac{1}{4x^2-1} \geq -1$:

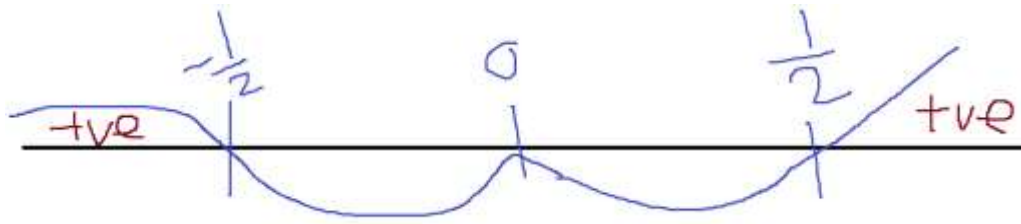
$$\begin{aligned}\frac{1}{4x^2-1} + \frac{4x^2-1}{4x^2-1} &\geq 0 \\ \frac{4x^2}{4x^2-1} &\geq 0\end{aligned}$$

Divide both sides by 4:

$$\begin{aligned}\frac{x^2}{\left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right)} &\geq 0 \\ \text{Roots} &\in \left\{-\frac{1}{2}, 0, \frac{1}{2}\right\}\end{aligned}$$

Check with -1 :

$$\frac{x^2}{\left(x + \frac{1}{2}\right)\left(x - \frac{1}{2}\right)} = \frac{(-1)^2}{\left(-1 + \frac{1}{2}\right)\left(-1 - \frac{1}{2}\right)} = \frac{+ve}{(-ve)(-ve)} = +ve$$

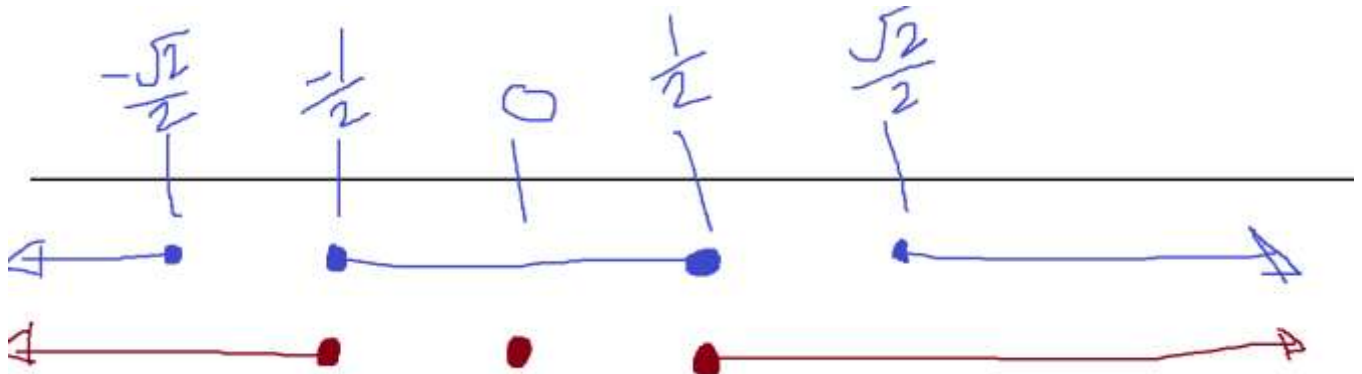


The solution set for the interval is

$$\text{Case II: } \left(-\infty, -\frac{1}{2}\right] \cup \{0\} \cup \left[\frac{1}{2}, \infty\right)$$

$$\text{Case I: } \left(-\infty, -\frac{\sqrt{2}}{2}\right] \cup \left[-\frac{1}{2}, \frac{1}{2}\right] \cup \left[\frac{\sqrt{2}}{2}, \infty\right)$$

$$\text{Zero Denominator not allowed: } \left\{-\frac{1}{2}, \frac{1}{2}\right\}$$



Take the intersection of Case I and II:

$$\left(-\infty, -\frac{\sqrt{2}}{2}\right] \cup \{0\} \cup \left[\frac{\sqrt{2}}{2}, \infty\right)$$

F. Greatest Integer Functions

Example 1.45

The domain of the function $f(x) = \sin^{-1}[2x^2 - 3] + \log_2\left(\log_{\frac{1}{2}}(x^2 - 5x + 5)\right)$ where $[t]$ is the greatest integer function is: (JEE-M 2022)

Domain of $\sin^{-1} x$

The input for $\sin^{-1} x$ must be between -1 and 1 :

$$-1 \leq [2x^2 - 3] \leq 1$$

Remove the greatest integer function:

$$-1 \leq 2x^2 - 3 < 2$$

$$2 \leq 2x^2 < 5$$

$$1 \leq x^2 < \frac{5}{2}$$

$$x^2 < \frac{5}{2} \Rightarrow -\sqrt{\frac{5}{2}} < x < \sqrt{\frac{5}{2}}$$

$$x^2 \geq 1 \Rightarrow x \geq 1 \text{ OR } x \leq -1$$

$$\left(-\frac{\sqrt{10}}{2}, -1\right] \cup \left[1, \frac{\sqrt{10}}{2}\right)$$

Domain of the log function

For the inner log function:

$$x^2 - 5x + 5 > 0$$

$$x = \frac{5 \pm \sqrt{25 - 4(1)(5)}}{2(1)} = \frac{5 \pm \sqrt{5}}{2}$$

$$\left(-\infty, \frac{5 - \sqrt{5}}{2}\right) \cup \left(\frac{5 + \sqrt{5}}{2}, \infty\right)$$

$$\frac{5 - \sqrt{5}}{2} \approx \frac{5 - 2.23}{2} = \frac{2.77}{2} = 1.39$$

$$\frac{5 + \sqrt{5}}{2} \approx \frac{5 + 2.23}{2} = \frac{7.23}{2} = 3.615$$

For the outer log function:

$$\log_{\frac{1}{2}}(x^2 - 5x + 5) > 0$$

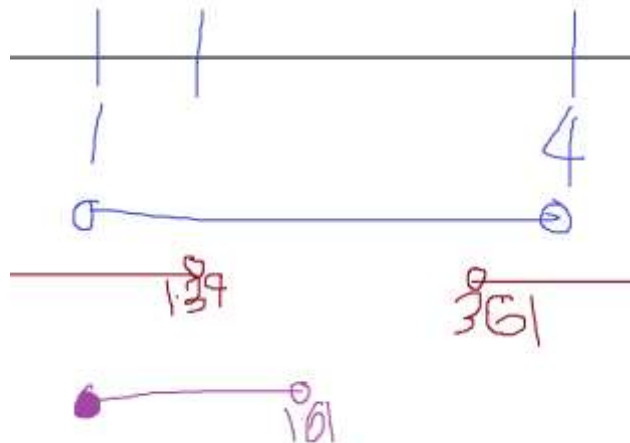
Using log properties, take the reciprocal of the base:

$$-\log_2(x^2 - 5x + 5) > 0$$

$$x^2 - 5x + 4 < 0$$

$$(x - 4)(x - 1) < 0$$

$$x \in (1, 4)$$



The final answer is:

$$x \in \left(1, \frac{5 - \sqrt{5}}{2}\right)$$

G. Absolute Value

Example 1.46

Determine the domain of the function: (JEE-M 2020)

$$f(x) = \sin^{-1} \left(\frac{|x| + 5}{x^2 + 1} \right)$$

H. $\sec^{-1} x$

1.47: Domain of $\sec^{-1} x$

The domain of $\sec^{-1} x$ is the range of $\sec x$

$$D_{\sec^{-1} x} = (-\infty, -1] \cup [1, \infty)$$

$$\text{Range}_{\cos x} \in [-1, 1]$$

$$\text{Range}_{\sec x} \in (-\infty, -1] \cup [1, \infty)$$

The domain of an inverse function is the range of the original function:

$$D_{\sec^{-1} x} = R_{\sec x} = (-\infty, -1] \cup [1, \infty)$$

Example 1.48

Determine the domain of the function $f(x) = \sec^{-1} \left(\frac{2x}{5x+3} \right)$ (JEE-M 2023, Adapted)

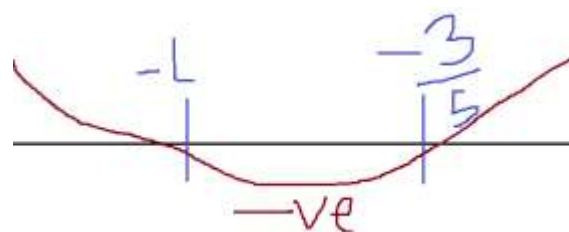
$$\frac{2x}{5x+3} \text{ has degree} = 1 - 1 = 0 \Rightarrow \text{Even}$$

$$\frac{2x}{5x+3} \geq 1 \Rightarrow \frac{-3x-3}{5x+3} \geq 0 \Rightarrow \frac{x+1}{5x+3} \leq 0$$

$$\text{Roots} \in \left\{ -1, -\frac{3}{5} \right\}$$

$$\left[-1, -\frac{3}{5} \right)$$

where $-\frac{3}{5}$ was excluded since it is a denominator root, and makes the expression undefined.

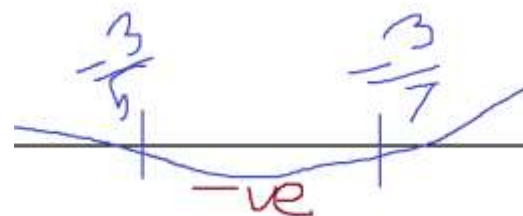


$$\frac{2x}{5x+3} \leq -1 \Rightarrow \frac{7x+3}{5x+3} \leq 0$$

$$\text{Roots} \in \left\{ -\frac{3}{5}, -\frac{3}{7} \right\}$$

$$\left(-\frac{3}{5}, -\frac{3}{7} \right]$$

where $-\frac{3}{5}$ was excluded since it is a denominator root, and makes the expression undefined.



$$\left[-1, -\frac{3}{5} \right) \cup \left(-\frac{3}{5}, -\frac{3}{7} \right]$$

I. Range

Example 1.49

The range of $f(x) = 4 \sin^{-1} \left(\frac{x^2}{x^2+1} \right)$ is: (JEE-M 2023)

Note that

$$0 < \frac{x^2}{x^2+1} < 1$$

Apply the $\sin$ inverse function throughout:

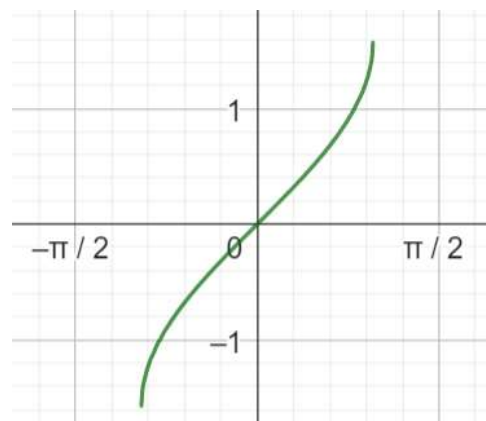
$$0 < \sin^{-1} \left(\frac{x^2}{x^2+1} \right) < \frac{\pi}{2}$$

Multiply by 4 throughout:

$$0 < 4 \sin^{-1} \left(\frac{x^2}{x^2+1} \right) < 2\pi$$

The range is

$$(0, 2\pi)$$



2. FURTHER TOPICS

2.1 Parametric Equations

A. Basics

2.1: Function

$y = f(x)$ is a function if there is rule such that for every valid x in the domain, there is an output y .

Some curves (such as circles) are not functions.

Some curves are difficult to represent using functions.

2.2: Parametric Representation

In a parametric representation, we describe a graph in the (x, y) coordinate using a third variable (usually called t). x and y are functions of t . That is

$$y = f(t), \quad x = f(t)$$

B. Circles

2.3: Equation of a Circle

The cartesian equation of a circle center at origin and radius r is:

$$x^2 + y^2 = r^2$$

Draw a circle with center at the origin, and radius r .

Let P be a point on the circle with coordinates

$$(x, y)$$

Using the distance formula, the distance between P and the origin is:

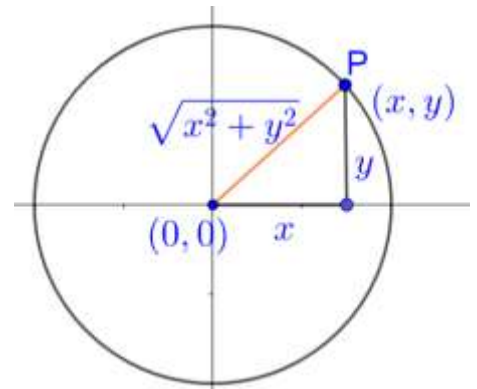
$$\sqrt{(x - 0)^2 + (y - 0)^2} = r$$

Simplify:

$$\sqrt{x^2 + y^2} = r$$

And finally, the equation of the circle is:

$$x^2 + y^2 = r^2$$



2.4: Pythagorean Identity

$$\sin^2 \theta + \cos^2 \theta = 1$$

The Pythagorean identity is useful in parameterization of circles and ellipses.

Example 2.5

Eliminate the parameter t from $x = \cos t, y = \sin t$ and determine the shape that it represents.

$$\begin{aligned} x^2 + y^2 &= \cos^2 t + \sin^2 t = 1 \\ x^2 + y^2 &= 1 \end{aligned}$$

This is the equation of the unit circle centered at origin.

Example 2.6

Eliminate the parameter t from the system of equations below and show that it represents a circle.

A. $x = r \cos t, y = r \sin t$

$$\begin{aligned}x^2 + y^2 &= r^2 \cos^2 t + r^2 \sin^2 t = r^2 \\x^2 + y^2 &= r^2\end{aligned}$$

This is the equation of a circle of radius r centred at origin.

2.7: Equation of a Circle

The equation of a circle with center (h, k) and radius r is:

$$(x - h)^2 + (y - k)^2 = r^2$$

Example 2.8

Eliminate the parameter t from the system of equations below and show that it represents a circle.

A. $x = 2 + 3 \cos t, y = 3 \sin t - 1$

$$\cos^2 t + \sin^2 t = 1$$

Substitute $\cos t = \frac{x-2}{3}, \sin t = \frac{y+1}{3}$:

$$\begin{aligned}\frac{(x-2)^2}{3^2} + \frac{(y+1)^2}{3^2} &= 1 \\(x-2)^2 + (y+1)^2 &= 3^2\end{aligned}$$

Circle of radius 3 centred at $(2, -1)$.

Example 2.9

Eliminate the parameter t from the system of equations below and show that it represents a circle.

A. $x = h + r \cos t, y = k + r \sin t$

$$(x - h)^2 + (y - k)^2 = r^2$$

$$\begin{aligned}x - h &= r \cos t \\y - k &= r \sin t\end{aligned}$$

$$LHS = (x - h)^2 + (y - k)^2 = (r \cos t)^2 + (r \sin t)^2 = r^2 = RHS$$

Substitute $\cos t = \frac{x-h}{r}, \sin t = \frac{y-k}{r}$ in the identity $\cos^2 t + \sin^2 t = 1$:

$$\frac{(x-h)^2}{r^2} + \frac{(y-k)^2}{r^2} = 1 \Rightarrow (x-h)^2 + (y-k)^2 = r^2$$

Circle of radius r centred at (h, k) .

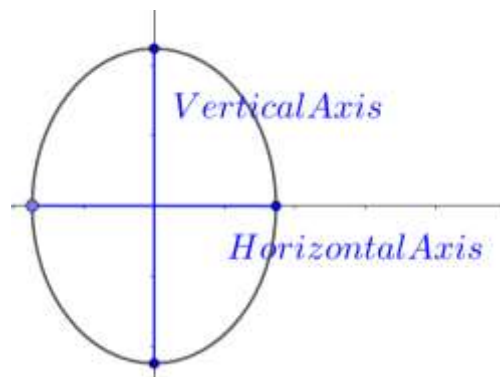
Example 2.10

Ferris Wheel

C. Ellipses**2.11: Equation of an Ellipse**

The cartesian equation of an ellipse centered at the origin is:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

A circle is a special case of an ellipse with $a = b$.**Example 2.12**Eliminate the parameter t from the system of equations below and identify the nature of the function it represents.

A. $x = \sqrt{5} \cos t, y = 3 \sin t$

$$\cos^2 t + \sin^2 t = 1 \Rightarrow \underbrace{\frac{a^2 \cos^2 t}{a^2} + \frac{b^2 \sin^2 t}{b^2}}_{\text{Equation I}} = 1$$

$$x = \sqrt{5} \cos t \Rightarrow x^2 = 5 \cos^2 t \Rightarrow \cos^2 t = \frac{x^2}{5}$$

$$y = 3 \sin t \Rightarrow y^2 = 9 \sin^2 t \Rightarrow \sin^2 t = \frac{y^2}{9}$$

Substitute $\cos^2 t = \frac{x^2}{5}, \sin^2 t = \frac{y^2}{9}$ in $\cos^2 t + \sin^2 t = 1$:

$$\frac{x^2}{5} + \frac{y^2}{9} = 1 \Rightarrow 9x^2 + 5y^2 = 45$$

2.13: Equation of an Ellipse

$$x = a \cos t, y = b \sin t \Leftrightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\begin{aligned} \cos^2 t + \sin^2 t &= 1 \\ \frac{a^2 \cos^2 t}{a^2} + \frac{b^2 \sin^2 t}{b^2} &= 1 \end{aligned}$$

Substitute $x = a \cos t, y = b \sin t$ in Equation I to get an ellipse centered at the origin:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

2.14: Equation of an EllipseThe equation of an ellipse centered at (h, k) with axes a and b is

$$\frac{(x-h)^2}{a^2} + \frac{(y-k)^2}{b^2} = 1$$

Example 2.15

Eliminate the parameter t from the system of equations below and identify the nature of the function it represents.

$$1. \quad x = h + a \cos t, y = k + b \sin t$$

Substitute:

$$x - h = a \cos t \Rightarrow (x - h)^2 = a^2 \cos^2 t$$

$$y - k = b \sin t \Rightarrow (y - k)^2 = b^2 \sin^2 t$$

In Equation 1 to get an ellipse centered at (h, k)

$$\frac{(x - h)^2}{a^2} + \frac{(y - k)^2}{b^2} = 1$$

D. Parabola

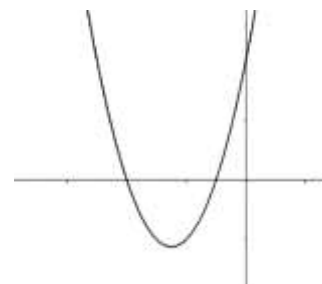
2.16: Parabola

$$y = ax^2 + bx + c$$

$$y = x^2$$

In a parabola, one quantity is related to the square of another quantity.

The parameter in one variable is the square of what it is in the other variable.



2.17: Algebraic Parametrization

2.18: Restrictions

Example 2.19

Eliminate the parameter t from the system of equations below and show it represents a parabola, or part of a parabola.

$$x = \sqrt{t}, \quad y = t$$

$$x = \sqrt{t} \Rightarrow x^2 = t$$

Substitute:

$$y = t \Rightarrow y = x^2$$

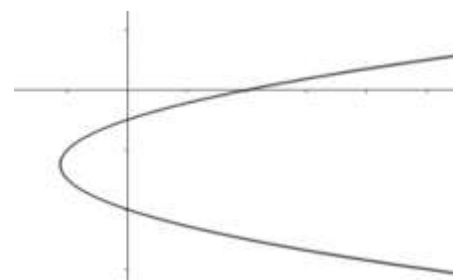
This is a parabola that only holds for

1. $x > 0$ since the square root function has nonnegative output:

$$\text{Range of } \sqrt{t} = [0, \infty)$$

2. $t > 0$ since the square root function has nonnegative domain:

$$\text{Domain of } \sqrt{t} = [0, \infty)$$



2.20: Parabola

$$x = ay^2 + by + c$$

Example 2.21

Eliminate the parameter t from the system of equations below and show it represents a parabola, or part of a

parabola.

$$x = t^2, \quad y = t + 2$$

$$y = t + 2 \Rightarrow t = y - 2$$

$$x = t^2 = (y - 2)^2 = y^2 - 4y + 4$$

Parabola with no restrictions

2.22: Exponential Parametrization

Example 2.23

Eliminate the parameter t from the system of equations below and show it represents a parabola, or part of a parabola.

$$x = e^t - 1, \quad y = e^{2t}$$

Method I

$$\begin{aligned} x &= e^t - 1 \\ e^t &= x + 1 \\ e^{2t} &= (x + 1)^2 \\ y &= (x + 1)^2 \end{aligned}$$

To establish the interval for x , note that:

$$\begin{aligned} x &= f(t) = e^t - 1 \\ e^t > 0 &\Rightarrow e^t - 1 > -1 \Rightarrow x > -1 \\ \text{Range is } &(-1, \infty) \end{aligned}$$

Method II

$$\begin{aligned} e^t &= x + 1 \\ t &= \ln(x + 1) \\ y &= e^{2t} = e^{2\ln(x+1)} = e^{\ln(x+1)^2} = (x + 1)^2 \end{aligned}$$

That is the right side of the parabola, not including its vertex.

2.24: Trigonometric Parametrization

Example 2.25

Eliminate the parameter t from the system of equations below and show it represents a parabola, or part of a parabola.

$$\text{A. } x = \sin t, y = \cos 2t, -\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$$

$$y = \cos 2t = 1 - 2 \sin^2 t = 1 - 2x^2$$

Over the domain $-\frac{\pi}{2} \leq t \leq \frac{\pi}{2}$, t has the range $[-1, 1]$, starting at -1 and going to 1 .

Hence, the parabola begins at $(-1, -1)$ and ends at $(1, -1)$

Example 2.26

Eliminate the parameter t from the system of equations below and identify the nature of the function it represents.

$$\begin{aligned} y^2 &= 4ax \\ y &= 2at \Rightarrow y^2 = 4a^2t^2 = (4a)(at^2) \\ x &= at^2 \end{aligned}$$

E. Hyperbola

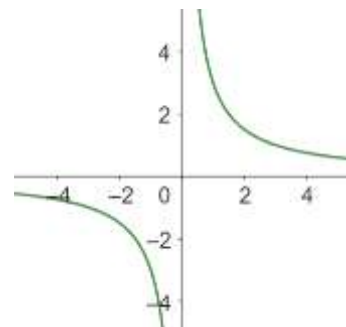
2.27: Equation of a Rectangular Hyperbola

The equation of a rectangular hyperbola is:

$$xy = k$$

Where

k is constant of proportionality



$$PV = \text{Constant}$$

$$\text{No. of Apples} \times \text{Price per Apple} = \text{Total Money}$$

Example 2.28

Eliminate the parameter t from the system of equations below and identify the nature of the function it represents.

- A. $x = \sin t, y = \csc t = \frac{1}{\sin t}$
- B. $x = \sqrt{k} \sin t, y = \sqrt{k} \csc t$
- C. $x = k \sin t, y = \csc t$
- D. $x = \sin t, y = k \csc t$

Part A

$$xy = (\sin t) \left(\frac{1}{\sin t} \right) = 1$$

Rectangular Hyperbola with constant of proportionality 1

Parts B, C and D

$$xy = k$$

Rectangular Hyperbola with constant of proportionality k

2.29: Equation of a Hyperbola

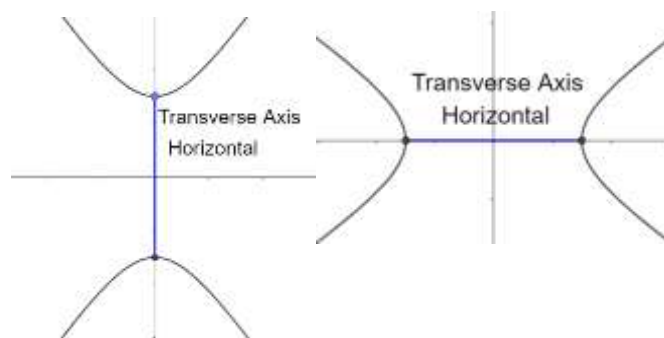
The equation of a hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = k, \quad \text{Transverse Axis Horizontal}$$

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = k, \quad \text{Transverse Axis Horizontal}$$

Compare with the equation of an ellipse:

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$



Example 2.30

Eliminate the parameter t from the system of equations below and identify the nature of the function it represents.

A. $x = t + \frac{1}{t}, y = t - \frac{1}{t}$
 B. $x = \sqrt{3 + t^2}, y = t$

Part A

From the given equations:

$$\underbrace{x + y = 2t}_{\text{Equation I}}, \underbrace{x - y = \frac{2}{t}}_{\text{Equation II}}$$

Multiply Equation I and II:

$$(x + y)(x - y) = (2t)\left(\frac{2}{t}\right)$$

$$x^2 - y^2 = 4$$

Example 2.31

Transverse Axis Horizontal: $x = a \sec t, y = b \tan t$

Transverse Axis Vertical: $x = b \tan t, y = a \sec t$

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = \frac{(a \sec t)^2}{a^2} - \frac{(b \tan t)^2}{b^2} = \frac{a^2 \sec^2 t}{a^2} - \frac{b^2 \tan^2 t}{b^2} = \sec^2 t - \tan^2 t = 1$$

$$\frac{y^2}{a^2} - \frac{x^2}{b^2} = \frac{(a \sec t)^2}{a^2} - \frac{(b \tan t)^2}{b^2} = \frac{a^2 \sec^2 t}{a^2} - \frac{b^2 \tan^2 t}{b^2} = \sec^2 t - \tan^2 t = 1$$

F. Lines**2.32: Line****Example 2.33**

Eliminate the parameter t from each system of equations below and identify the nature of the function it represents.

A. $x = 3 + t, y = 2 + 5t$
 B. $x = a + t, y = b + mt$

Part A

Substitute $x = 3 + t \Rightarrow t = x - 3$ in the second equation:

$$y = 2 + 5t = 2 + 5(x - 3) = 5x - 13$$

This is a line with slope 5, and y-intercept -13.

Part A

Substitute $x = a + t \Rightarrow t = x - a$ in the second equation:

$$y = b + mt = b + m(x - a) = mx + b - ma$$

This is a line with slope m , and y-intercept $b - a$.

Example 2.34

Convert the equation $y = 4x + 7$ into parametric form.

$$t = x \Rightarrow y = 4t + 7$$

$$t = x + 2 \Rightarrow x = t - 2 \Rightarrow y = 4t + 7$$

G. Further Questions

Example 2.35

Show that $x^2y + a^2y = abx$, $ab > 0$ (Newton's Serpentine) can be represented parametrically as:

$$x = a \cot t, \quad y = b \sin t \cos t$$

$$x^2y + a^2y - abx = 0$$

Substitute $x = a \cot t$, $y = b \sin t \cos t$ in the LHS:

$$(a^2 \cot^2 t)(b \sin t \cos t) + a^2(b \sin t \cos t) - ab(a \cot t)$$

Factor out $a^2b \cot t$:

$$(a^2b \cot t)[(\cot t)(\sin t \cos t) + (\sin t \cos t \tan t) - 1]$$

Simplify:

$$\begin{aligned} & (a^2b \cot t)[\cos^2 t + \sin^2 t - 1] \\ & (a^2b \cot t)[1 - 1] = (a^2b \cot t)[0] = 0 = RHS \end{aligned}$$

Example 2.36

Eliminate the parameter θ from the system of equations below and solve for y :

$$x = 2 + \cos \theta \text{ for } 1 < x < 3, \quad y = 2 \csc \theta \text{ for } y > 2$$

$$\begin{aligned} \frac{2}{\sin \theta} = y & \Rightarrow \sin \theta = \frac{2}{y} \Rightarrow \sin^2 \theta = \frac{4}{y^2} \\ & \text{Equation I} \\ \cos \theta = x - 2 & \Rightarrow \cos^2 \theta = x^2 - 4x + 4 \\ & \text{Equation II} \end{aligned}$$

Add Equations I and II and substitute $\sin^2 \theta + \cos^2 \theta = 1$:

$$\begin{aligned} 1 &= \frac{4}{y^2} + x^2 - 4x + 4 \\ \frac{4}{y^2} &= -(x^2 - 4x + 3) \end{aligned}$$

Take the reciprocal of both sides. Then, multiply both sides by 4, take the square root both sides, and reject the negative square root:

$$\frac{y^2}{4} = -\frac{1}{x^2 - 4x + 3} \Rightarrow y = \frac{2}{\sqrt{-(x^2 - 4x + 3)}}$$

Check the domain:

$$-(x^2 - 4x + 3) > 0 \Rightarrow x^2 - 4x + 3 < 0 \Rightarrow (x - 3)(x - 1) < 0 \Rightarrow x \in (1, 3)$$

Check the range:

For $x \in (1, 3)$, $-(x^2 - 4x + 3)$ is a downward parabola with vertex $(2, 2)$. It takes values in the range $(0, 2)$ and hence $y = \frac{2}{\sqrt{-(x^2 - 4x + 3)}}$ takes values > 2 .

2.2 Polar Coordinates; Circles

A. Directed Distance

The rectangular coordinate system uses a grid layout to identify points. An alternate system specifies the location of a point as a combination of:

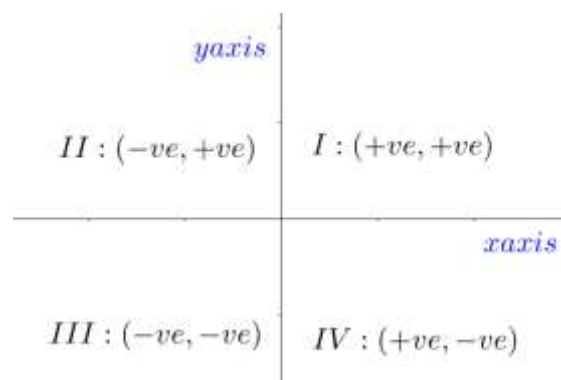
- Distance from the origin
- Angle in which the distance is to be travelled

Polar coordinates get used extensively in:

- Vectors
- Complex Numbers
- Calculus

2.37: Cartesian Coordinates

- The Cartesian coordinate plane consists of an x axis and a y axis perpendicular to the x axis.
- The axes divide the plane into quadrants.
- Any point on the plane is uniquely identified by its coordinates
- Any coordinates uniquely identify a point on the plane
- Cartesian coordinates are also called rectangular coordinates



2.38: Distance on the Real Number Line

Points a distance d on the real number line from the origin are given by:

$$|x| = d$$



$$|x| = 3 \Rightarrow x = \pm 3$$

Example 2.39

Identify the point on the real number line:

- A. 3 units to the right of zero
- B. 3 units to the left of zero
- C. -3 units to the right of zero
- D. -3 units to the left of zero

3 units to the right of zero = 3
 3 units to the left of zero = -3
 -3 units to the right of zero = -3
 -3 units to the left of zero = 3

2.40: Directed Distance = r

The directed distance r is defined as the distance travelled from the origin in a straight line in the direction indicated.

- $r > 0$ means you travel along the direction indicated
- $r < 0$ means you travel $|r|$ unit diametrically opposite (180° turn) along the direction indicated

Example 2.41

Identify the point that lies on the line $y = x$:

- A. That is reached by travelling 3 units at an angle of 45° with the positive x axis.
- B. That is reached by travelling 3 units at an angle of 225° with the positive x axis.
- C. That is reached by travelling -3 units at an angle of 225° with the positive x axis.

$$\begin{aligned} A: \left(\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}} \right) &= \left(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2} \right) \\ B: \left(-\frac{3}{\sqrt{2}}, -\frac{3}{\sqrt{2}} \right) &= \left(-\frac{3}{2}\sqrt{2}, -\frac{3}{2}\sqrt{2} \right) \\ C: \left(\frac{3}{\sqrt{2}}, \frac{3}{\sqrt{2}} \right) &= \left(\frac{3}{2}\sqrt{2}, \frac{3}{2}\sqrt{2} \right) \end{aligned}$$

B. Polar Coordinates**2.42: Polar Coordinates**

Polar coordinates define a point in the form:

$$(r, \theta)$$

Where

$r = \text{directed distance}$

$\theta = \text{angle made by the point with polar axis}$

$\text{Polar axis} = \text{positive direction of } x \text{ axis}$

Example 2.43

Convert from polar coordinates to Cartesian Coordinates:

- A. $\left(3, \frac{\pi}{3} \right)$
- B. $\left(2, \frac{\pi}{4} \right)$

$$\underbrace{\left(3, \frac{\pi}{3} \right)}_{\text{Polar Form}} = \underbrace{\left(\frac{3}{2}, \frac{3\sqrt{3}}{2} \right)}_{\text{Rectangular Form}}$$

$$\underbrace{\left(2, \frac{\pi}{4} \right)}_{\text{Polar Form}} = \underbrace{(\sqrt{2}, \sqrt{2})}_{\text{Rectangular Form}}$$

2.44: Different Values of θ

$$(r, \theta) = (r, \theta + n360^\circ) = (r, \theta + 2\pi n), \quad n \in \mathbb{Z}$$

- There are an infinite number of values of θ that will result in the same angle.
- From basic trigonometry, we know that adding a full circle does not change the angle of the final result.

2.45: Converting from positive to negative r

$$(r, \theta) = (-r, \theta + \pi) = (-r, \theta - \pi)$$

Example 2.46

Convert to rectangular form:

- A. $\left(-3, \frac{\pi}{4}\right)$
- B. $\left(2, -\frac{\pi}{4}\right)$
- C. $\left(-2, -\frac{\pi}{4}\right)$

Part A

$$\left(-3, \frac{\pi}{4}\right) = (-3, 45^\circ)$$

Use the property $(r, \theta) = (-r, \theta + \pi)$:

$$(3, 45^\circ + 180^\circ) = (3, 225^\circ) = \left(-\frac{3\sqrt{2}}{2}, -\frac{3\sqrt{2}}{2}\right)$$

Part B

$$\left(2, -\frac{\pi}{4}\right) = (2, -45^\circ) = \underbrace{(\sqrt{2}, -\sqrt{2})}_{\text{Rectangular Form}}$$

Part C

$$\left(-2, -\frac{\pi}{4}\right) = (2, -45^\circ + 180^\circ) = (2, 135^\circ) = \underbrace{(-\sqrt{2}, \sqrt{2})}_{\text{Rectangular Form}}$$

Example 2.47

Given that $|\theta| < 2\pi$, Write $(1, \sqrt{3})$ in polar form (r, θ) with:

- A. $r > 0, \theta > 0$
- B. $r < 0, \theta > 0$
- C. $r > 0, \theta < 0$
- D. $r < 0, \theta < 0$

Combine the answers from all parts and compare.

The absolute value can be removed to get the equivalent restriction for the range of θ :

$$-2\pi < \theta < 2\pi$$

The distance from the origin

$$= r = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{4} = 2$$

Part A

Since we get a 30 – 60 – 90 triangle the angle is

$$\theta = 60^\circ$$

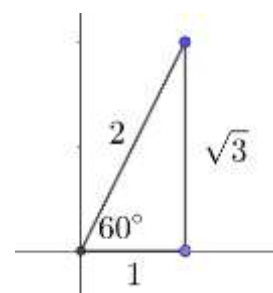
$$(r, \theta) = (2, 60^\circ) = \left(2, \frac{\pi}{3}\right)$$

Part B

Using the property $(r, \theta) = (-r, \theta + \pi)$:

$$\left(2, \frac{\pi}{3}\right) = \left(-2, \frac{\pi}{3} + \pi\right) = \left(-2, \frac{4\pi}{3}\right)$$

Part C



We replace $\theta = \frac{\pi}{3}$ with an equivalent negative angle

$$\left(2, \frac{\pi}{3}\right) = \left(2, \frac{\pi}{3} - 2\pi\right) = \left(2, -\frac{5\pi}{3}\right)$$

Part D

Modify the answer from Part A:

$$\left(2, \frac{\pi}{3}\right) = \left(-2, \frac{\pi}{3} - \pi\right) = \left(-2, -\frac{2\pi}{3}\right)$$

Or we can modify the answer from Part B:

$$\left(-2, \frac{4\pi}{3}\right) = \left(-2, \frac{4\pi}{3} - 2\pi\right) = \left(-2, -\frac{2\pi}{3}\right)$$

Writing all four answers together:

$$\underbrace{(1, \sqrt{3})}_{\text{Rectangular Form}} = \left(2, \frac{\pi}{3}\right) = \left(-2, \frac{4\pi}{3}\right) = \left(2, -\frac{5\pi}{3}\right) = \left(-2, -\frac{2\pi}{3}\right)$$

C. Equation of a Circle

In rectangular form, you would have encountered the equation of a line in multiple forms. They represent the same shape, but the form of the equation is different.

$$\underbrace{y = mx + c}_{\text{Slope Intercept Form}} \Leftrightarrow \underbrace{y - y_1 = m(x - x_1)}_{\text{Point Slope Form}}$$

$$\begin{aligned} y - y_1 &= mx - mx_1 \\ y &= mx - mx_1 + y_1 \end{aligned}$$

Substitute $c = -mx_1 + y_1$:

$$y = mx + c$$

Similarly, equations in polar coordinates can have more than one form.

2.48: Circle at Origin

A circle with center at the origin and radius R is the set of points that are exactly R units away from the origin.

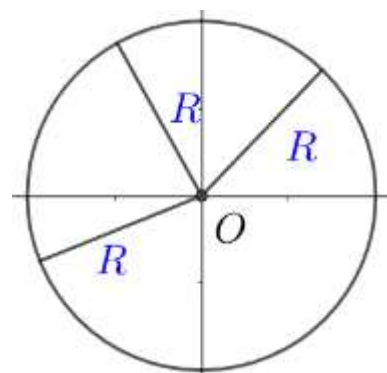
2.49: Circle

The equation of a circle in polar coordinates is given by:

$$r = R$$

Where:

r is a variable (directed distance)
 R is the radius of the circle



Example 2.50

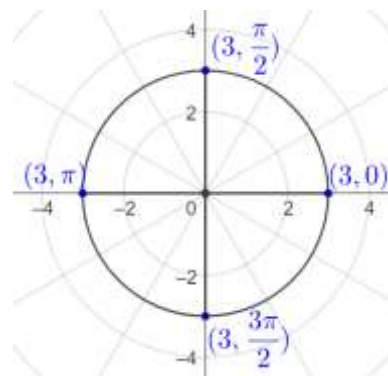
- Determine the shape formed by $r = 3$.
- Graph this shape. Make sure to identify the points where $\theta \in \left\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\}$ and write them in rectangular form and in polar form.

Part A

Circle with radius 3 and center at the origin

Part B

$$\begin{aligned}\theta = 0 &\Rightarrow P_1(3,0) \\ \theta = \frac{\pi}{2} &\Rightarrow P_2(3,0) \\ \theta = \pi &\Rightarrow P_3(-3,0) \\ \theta = \frac{3\pi}{2} &\Rightarrow P_4(0,-3)\end{aligned}$$



2.51: Circle

The equation of a circle in polar coordinates is given by:

$$r = -R$$

Where

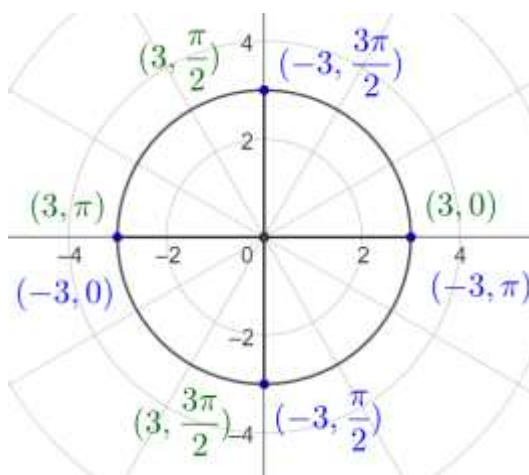
R is the radius of the circle

$$(r, \theta) = (-r, \theta + 180^\circ)$$

Since there is no restriction on the angle, even if we take r to be negative, the entire circle is covered.

Example 2.52

- Determine the shape formed by $r = -3$.
- Graph this shape. Make sure to identify the points where $\theta \in \{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\}$



Example 2.53

X travels at $10 \frac{km}{hr}$. Y travels at $5 \frac{km}{hr}$. Y is 5 km away from X. Y travels away X. At the same time, X goes to catch Y. When will X catch Y, and how much time will it take?

*Time: 1 Hour
Distance for Y: 5 km
Distance for X: 10 km*

$$Time = \frac{Distance}{Speed}$$

$$Time = \frac{Gap}{Speed} = \frac{5}{10 - 5} = \frac{5}{5} = 1$$

Example 2.54

Spaceship X travels along the curve $r = 3, t = \theta$. Spaceship Y travels along the curve $r = -3, t = \frac{\theta}{2}$. t represents time in minutes and $0 \leq t \leq 2\pi$.

- Write θ as a function of t for each of the spaceships.
- Determine if and when the paths of the two spaceships intersect.

Part A

For Spaceship X , angular Speed

$$= \text{Angle travelled per unit time} = 1 \frac{\text{rad}}{\text{min}}$$

For Spaceship Y , angular Speed

$$= \text{Angle travelled per unit time} = 2 \frac{\text{rad}}{\text{min}}$$

Part B

Starting point is:

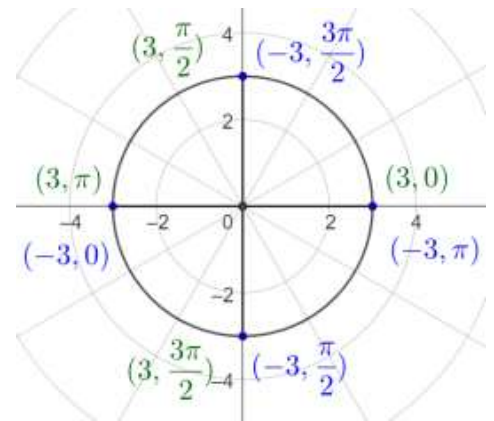
$$X = (3, 0), Y = (-3, 0)$$

The relative speed is:

$$2 - 1 = 1$$

The time taken by X to catch up with Y is

$$Time = \frac{Angle}{Angular Speed} = \frac{\pi}{2 - 1} = \pi$$



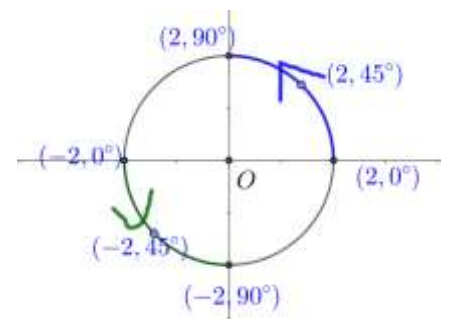
Example 2.55

Explain the difference in the circle $r = 2$ and the circle $r = -2$ by considering the restriction $0 \leq \theta \leq 90^\circ$

$r = 2$ begins at $(2, 0)$ and proceeds counterclockwise.

$r = -2$ begins at $(-2, -1)$ and proceeds counterclockwise.

In general, for any value of θ , $r = 2$ and $r = -2$ are at diametrically opposite points on the circle.

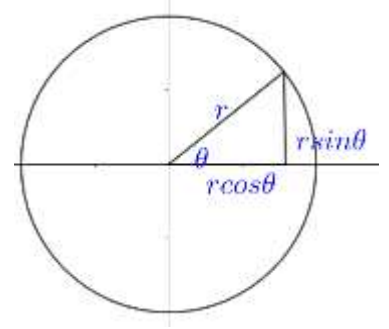


2.56: Polar to Cartesian Coordinates

Polar coordinates can be converted into Cartesian coordinates by using the equations:

$$x = r \cos \theta, y = r \sin \theta$$

$$\begin{aligned} x^2 + y^2 &= r^2 \cos^2 \theta + r^2 \sin^2 \theta = r^2 \\ x^2 + y^2 &= r^2 \end{aligned}$$

**2.57: Cartesian to Polar Coordinates**

The equation of a circle with center at the origin is given in Cartesian Coordinates by $x^2 + y^2 = R^2$. We can convert this into polar coordinates to show that

$$r = \pm R$$

$$\begin{aligned} x^2 + y^2 &= R^2 \\ (r \cos \theta)^2 + (r \sin \theta)^2 &= R^2 \\ r^2 \cos^2 \theta + r^2 \sin^2 \theta &= R^2 \\ r^2 (\cos^2 \theta + \sin^2 \theta) &= R^2 \\ r^2 &= R^2 \\ r &= \pm R \end{aligned}$$

2.58: [Equation](#) of a Circle: Alternate Form

Type equation here.

Multiply $r = 2a \cos \theta + 2b \sin \theta$ by r on both sides:

$$r^2 = 2ar \cos \theta + 2br \sin \theta$$

Convert to rectangular coordinates:

$$x^2 + y^2 = 2ax + 2by$$

Collate all terms to one side and complete the square:

$$x^2 - 2ax + y^2 - 2by = 0$$

$$(x - a)^2 + (y - b)^2 = a^2 + b^2$$

Voilà, circle of radius $\sqrt{a^2 + b^2}$ centered at (a, b) .

Example 2.59

The equation of a circle touching the origin is given in Cartesian coordinates by:

$$(x - 1)^2 + (y - 1)^2 = 4$$

Convert the equation into polar coordinates.

Expand the brackets:

$$x^2 - 2x + 1 + y^2 - 2y + 1 = 4$$

Rearrange and subtract 2 from both sides:

$$x^2 + y^2 - 2(x + y) = 2$$

Substitute $x^2 + y^2 = r^2$, $x = r \cos \theta$, $y = r \sin \theta$:

$$r^2 - 2r(\cos \theta + \sin \theta) = 2$$

$$r[1 - 2(\cos \theta + \sin \theta)] = 2$$

$$r = \frac{2}{1 - 2(\cos \theta + \sin \theta)}$$

Example 2.60

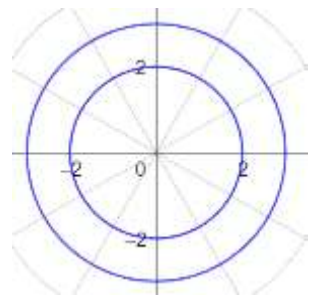
Graph the polar curve:

$$r^2 - 5r + 6 = 0$$

This is a quadratic. Factor and solve:

$$(r - 2)(r - 3) = 0 \Rightarrow r \in \{2, 3\}$$

This corresponds to 2 distinct circles with radii 2 and 3.



Example 2.61

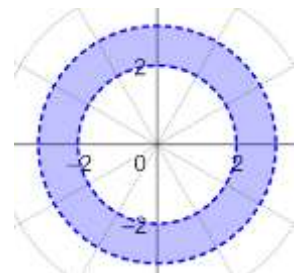
Graph the region that satisfies:

$$r^2 - 5r + 6 < 0, \quad r \in \mathbb{R}$$

$$(r - 2)(r - 3) < 0$$

$$2 < r < 3$$

And the region is the shaded area between the circles with radii 2 and 3 (graphed alongside).

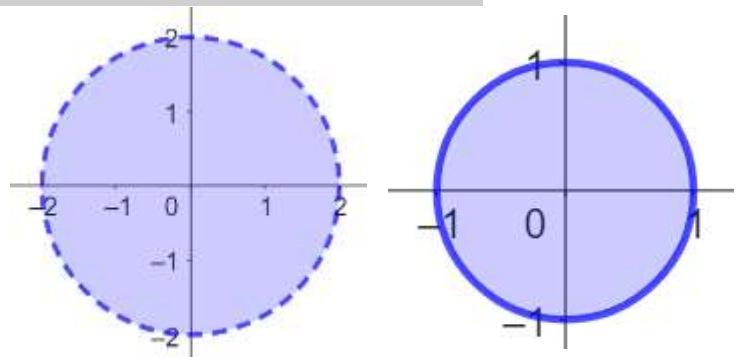


2.62: Disc

A circular disc in the coordinate plane with center at origin and radius R is given by

Boundary Included: $r \leq R$

Boundary Excluded: $r < R$



Example 2.63

The diagram alongside shows two circles. Write the equation of the circles in polar form.

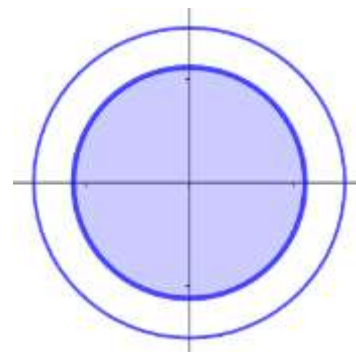
Left Circle: $r < 2$

Right Circle: $r \leq 1$

2.64: Annulus

Annulus is the region between two concentric circles.

White shaded region in the adjoining diagram.



2.65: Inequality for an Annulus

$$R_1 \leq r \leq R_2$$

Where

R_1 = Radius of inner circle

R_2 = Radius of outer circle

Endpoints included versus not included:

$$(r - R_1)(r - R_2) \leq 0$$

D. Equation of a Line

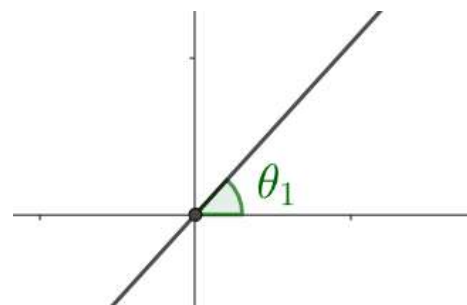
2.66: Equation of a Line

The equation of a line in polar coordinates going through the origin is given by

$$\theta = \theta_1$$

Where

θ_1 = angle made by line with x axis



Since there is no restriction on r

$$r \in (-\infty, \infty)$$

Example 2.67

Determine the equation of a line that passes the origin and:

A. (0,3)

B. $(-\sqrt{2}, -\sqrt{2})$

Part A

(0,3) lies on the y axis. Angle made by (0,3) with the x axis is 90° .

$$\theta = 90^\circ \text{ Or } \theta = \frac{\pi}{2}$$

Part B

$$(-\sqrt{2}, -\sqrt{2}) = (r, 225^\circ)$$

$$\theta = 225^\circ \text{ OR } \theta = \frac{5\pi}{4}$$

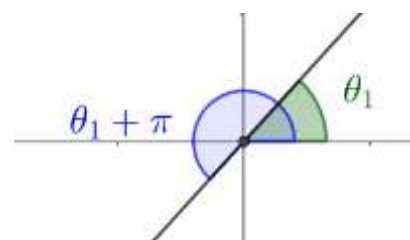
2.68: Equation of a Line-II

If $\theta = \theta_1$ is the equation of a line, then the equation is also represented by:

$$\theta = \theta_1 + 180^\circ = \theta_1 + \pi$$

Turning a line around by 180° results in the same line.

Hence, both equations are equally valid.



2.69: Converting from Rectangular Equations to Polar Equations

To convert from rectangular equations to polar coordinates use the following substitutions:

$$\begin{aligned}x &= r \cos \theta \\y &= r \sin \theta\end{aligned}$$

2.70: Horizontal and Vertical Lines

We can convert lines from Cartesian coordinates into lines in polar coordinates:

$$x = c \Rightarrow r = \frac{c}{\cos \theta}, \quad y = c \Rightarrow r = \frac{c}{\sin \theta}$$

Where

$$c \text{ is any constant, } c \in \mathbb{R}$$

Vertical Line

$$x = c \Rightarrow r \cos \theta = c \Rightarrow r = \frac{c}{\cos \theta}$$

Horizontal Line

$$y = c \Rightarrow r \sin \theta = c \Rightarrow r = \frac{c}{\sin \theta}$$

Example 2.71

Write the following Cartesian equations in polar form:

- A. $x = 3$
- B. $y = 2$

$$\begin{aligned}r \cos \theta = 3 &\Rightarrow r = \frac{3}{\cos \theta} \\r \sin \theta = 2 &\Rightarrow r = \frac{2}{\sin \theta}\end{aligned}$$

2.72: General Equation of a Line

The equation of a line that does not pass through the origin:

$$r = \frac{c}{\sin \theta - m \cos \theta}$$

$$y = mx + c \Rightarrow y - mx = c$$

Substitute $y = r \sin \theta$, $x = r \cos \theta$:

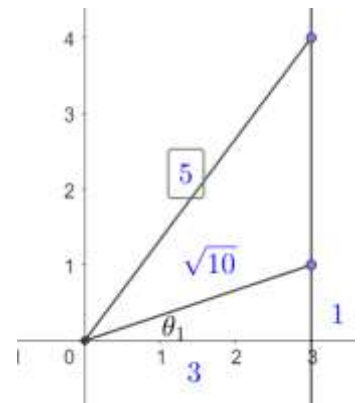
$$r \sin \theta - m r \cos \theta = c$$

Factor r :

$$r(\sin \theta - m \cos \theta) = c$$

Solve for r :

$$r = \frac{c}{\sin \theta - m \cos \theta}$$



E. Inequalities

Just as equations lead to shapes in the coordinate, inequalities represent intervals on the number line, and lead to regions in the coordinate plane.

2.73: Region between two lines

Example 2.74

Describe the region between the lines $y = x$ and $y = \frac{x}{\sqrt{3}}$ using polar coordinates. (Assume the region includes its boundaries).

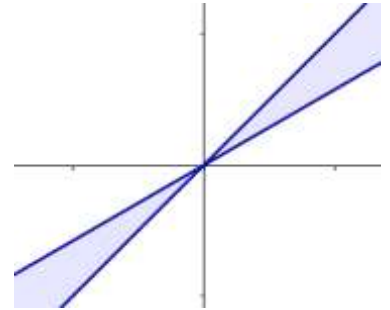
$$y = x \Rightarrow \theta = 45^\circ$$

$$y = \frac{x}{\sqrt{3}} \Rightarrow \theta = 30^\circ$$

The region that we want is:

$$30^\circ \leq \theta \leq 45^\circ$$

This region has infinite area.



Converting from Polar to Cartesian and vice versa

$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$r^2 = x^2 + y^2$$

$$\tan \theta = y/x$$

Example 2.75

Convert from Polar to Cartesian

- A. $r \sin \theta = 5$
- B. $r \cos \theta = 7$
- C. $r^2 \sin 2\theta = 1$
- D. $r^2 \sin^2 \theta = 1$

Parts A and B

$y=5$, Equation of horizontal line

$x=7$, Equation of vertical line

Part C

$$r^2 \sin 2\theta = 1$$

Substitute $2 \sin \theta \cos \theta = \sin 2\theta$

$$2r^2 \sin \theta \cos \theta = 1$$

$$(r \sin \theta)(r \cos \theta) = \frac{1}{2}$$

$$xy = \frac{1}{2}$$

Rectangular Hyperbola

Part D

$$r^2 \sin^2 \theta = 1$$

$$(r \sin \theta)^2 = 1$$

$$y^2 = 1$$

$$y = \pm 1$$

Part E

$$r^2 \cos^2 \theta - r^2 \sin^2 \theta = 1$$

$$x^2 - y^2 = 1$$

Hyperbola

Example 2.76

Convert from Cartesian to Polar. Identify the equation.

A. $x^2 + y^2 = c$

B.

C. $r = \frac{3}{4 \cos \theta - 2 \sin \theta}$

Part A

Convert from Cartesian to Polar. Identify it is as standard equation

$$x^2 + y^2 = c$$

$$r = \sqrt{c}$$

Part B

Convert from Polar to Cartesian: Circle

$$r = a \cos \theta$$

$$r^2 = a r \cos \theta$$

$$x^2 + y^2 = ax$$

$$x^2 - ax + y^2 = 0$$

$$x^2 - ax + a^2/4 + y^2 = a^2/4$$

$$(x - a/4)^2 + y^2 = a^2/4$$

Part C

Eliminate fractions:

$$4r \cos \theta - 2r \sin \theta = 3$$

$$4x - 2y = 3$$

$$y = 2x - \frac{3}{2}$$

2.3 Graphing in Polar Coordinates

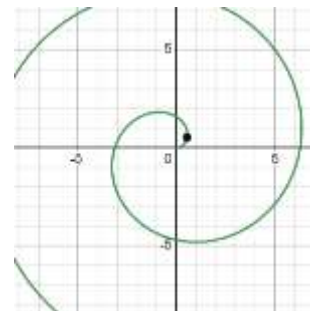
A. Archimedes Spirals

There are certain “named” polar curves, which you will need to remember.

2.77: Graphing Polar Equations

To graph polar equations, we make a table of values.

When the values start to repeat, we stop the graph.



2.78: Archimedes Spiral

The polar curve where the distance from the origin is given by the angle itself is the equation:

$$r = \theta$$

Variants of the

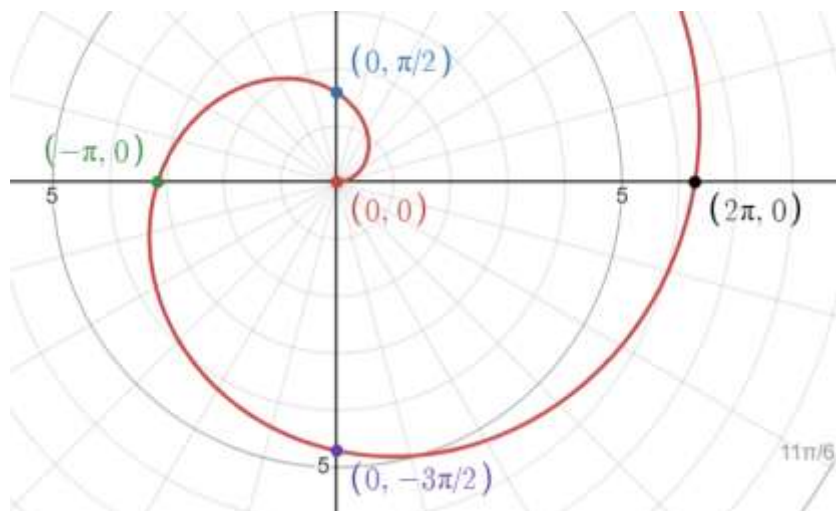


Example 2.79

Draw the polar equation $r = \theta$ and plot the points

$$r \in \left\{0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}\right\}$$

The graph is plotted, and the points are plotted with rectangular coordinates shown:



B. Circles

2.80: Circles

A circle of radius R centered at the origin has the equation:

$$r = R$$

$$r = -R$$

2.81: Circles: Trigonometric Version

A circle with diameter a and center at the origin has the equation

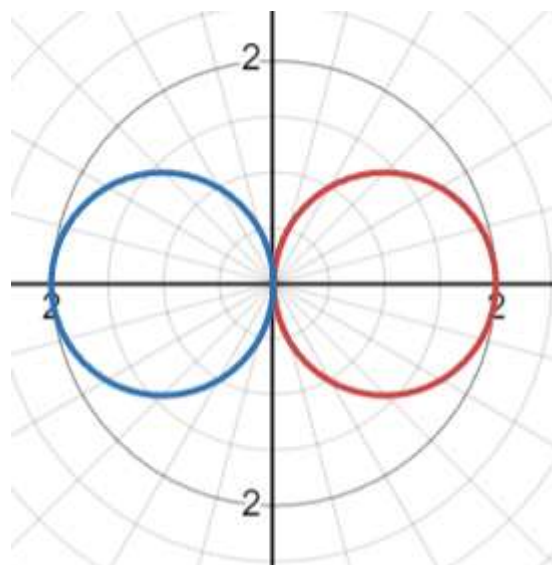
$$r = a \cos \theta$$

- Oriented "right" of the origin: if $a > 0$
- Oriented "left" of the origin: if $a < 0$

The graph alongside has

$$r = 2 \cos \theta \text{ in red}$$

$$r = -2 \cos \theta \text{ in blue}$$



Example 2.82

$$r = 2 \cos \theta$$

$$x = r \cos \theta = (a \cos \theta) \cos \theta = a \cos^2 \theta$$

$$y = r \sin \theta = (a \cos \theta) \sin \theta$$

When $\theta = 0$:

$$(x, y) = (2 \cos^2 0, 2 \cos 0 \sin 0) = (2, 0)$$

When $\theta = \frac{\pi}{4}$:

$$(x, y) = \left(2 \cos^2 \frac{\pi}{4}, 2 \cos \frac{\pi}{4} \sin \frac{\pi}{4}\right) = \left(2 \left(\frac{1}{\sqrt{2}}\right)^2, 2 \cdot \frac{1}{\sqrt{2}} \cdot \frac{1}{\sqrt{2}}\right) = (1, 1)$$

When $\theta = \frac{\pi}{2}$:

$$(x, y) = \left(2 \cos^2 \frac{\pi}{2}, 2 \cos \frac{\pi}{2} \sin \frac{\pi}{2}\right) = (2 \cdot 0^2, 2 \cdot 0 \cdot 1) = (0, 0)$$

When $\theta = \pi$:

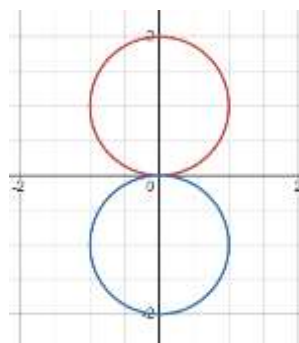
$$(x, y) = (2 \cos^2 \pi, 2 \cos \pi \sin \pi) = (2 \cdot (-1)^2, 2 \cdot -1 \cdot 0) = (2, 0)$$

2.83: Circles: Trigonometric Version

A circle with diameter a and center at the origin has the equation

$$r = a \sin \theta$$

- Oriented "top": if $a > 0$
- Oriented "bottom": if $a < 0$



The graph alongside has

θ	$r = 2 \sin \theta$ in red	$r = -2 \sin \theta$ in blue
0	0	0
$\frac{\pi}{4}$	$\sqrt{2}$	$-\sqrt{2}$
$\frac{\pi}{2}$	2	-2
$\frac{3\pi}{4}$	$\sqrt{2}$	$-\sqrt{2}$
π	0	0

C. Limacons

2.84: Limacons

Limacons are an important class of curves. They have the form:

$$r = a \pm b \cos \theta$$

$$r = a \pm b \sin \theta$$

$$a > 0, b > 0$$

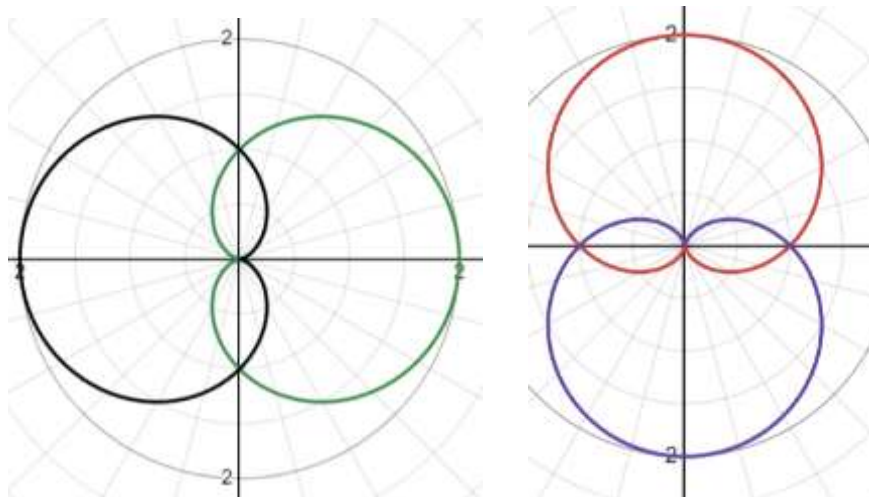
2.85: Cardoid

A limaçon is a cardoid if $\frac{a}{b} = 1$. That is

$$a = b$$

If $r = a \pm b \sin \theta$, the limaçon is upward oriented if

Upward oriented: b has $a + ve$ sign $\Rightarrow$ Red graph
 Downward oriented: b has $a - ve$ sign $\Rightarrow$ Violet graph



If $r = a \pm b \cos \theta$, the limaçon is upward oriented if

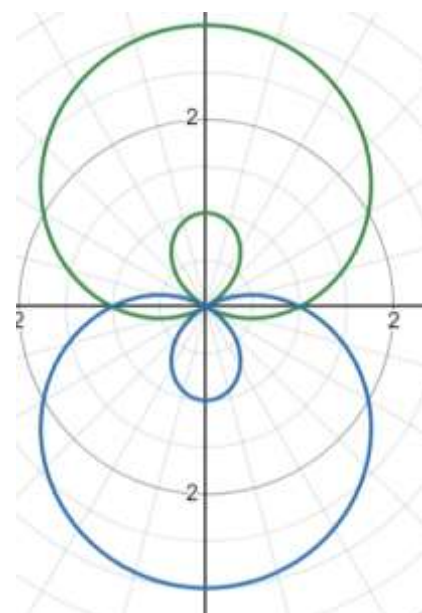
Rightward oriented: b has $a + ve$ sign $\Rightarrow$ Green graph
 Leftward oriented: b has $a - ve$ sign $\Rightarrow$ Black graph

2.86: Inner Loop Limaçons

A limaçon has an inner loop if

$$\frac{a}{b} < 1 \Rightarrow a < b$$

Upward Oriented: $r = 1 + 2 \sin \theta$: Green Graph
 Downward Oriented: $r = 1 - 2 \sin \theta$: Blue Graph



2.87: One Loop Limaçons

A limaçon has one loop if

$$1 < \frac{a}{b} < 2$$

This is also called a *dimpled* limaçon.

$$a = 4, b = 3$$

2.88: Simple Closed Limacons

A simple closed limaçon has

$$\frac{a}{b} > 2$$

Example 2.89

Determine when does a limaçon pass through the origin. Also, determine the values of θ at which this occurs.

$$r = a + b \cos \theta = 0$$

$$\cos \theta = -\frac{a}{b}$$

$$\theta = \cos^{-1}\left(\frac{a}{b}\right)$$

This is valid if

$$-1 \leq \frac{a}{b} \leq 1$$

Since $a > 0, b > 0$ we can drop the negative condition:

$$\frac{a}{b} \leq 1$$

$$a \leq b$$

True for Cardioids and Inner Loop Limacons

Example 2.90

Determine the maximum and minimum distance from the origin for $r(\theta) = a + b \cos \theta, \frac{a}{b} > 1$. Also determine the values of θ at which this occurs.

a is a constant

$$-1 < \cos \theta < 1$$

The graph will have maximum distance from the origin at

$$\theta = 0 + 2k\pi \Rightarrow \cos \theta = 1 \Rightarrow r = a + b \cos \theta = a + b(1) = a + b$$

The graph will have minimum distance from the origin at

$$\theta = \pi + 2k\pi \Rightarrow \cos \theta = -1 \Rightarrow r = a + b \cos \theta = a + b(-1) = a - b$$

D. Lemniscates

2.91: Lemniscates

This is a polar curve that resembles the infinity symbol. It is centered at the origin.

$$r^2 = a^2 \cos^2 \theta$$

$$r^2 = a^2 \sin^2 \theta$$

$$a \neq 0$$

E. Roses

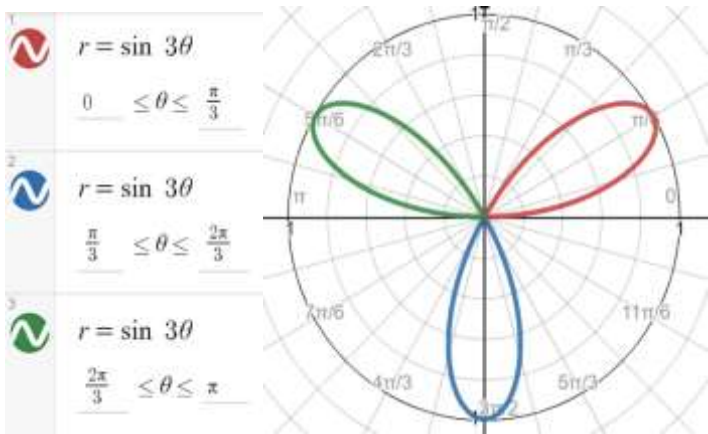
2.92: Graphing Roses

$$r = \sin n\theta, n \text{ is even} \Rightarrow 2n \text{ petals}$$

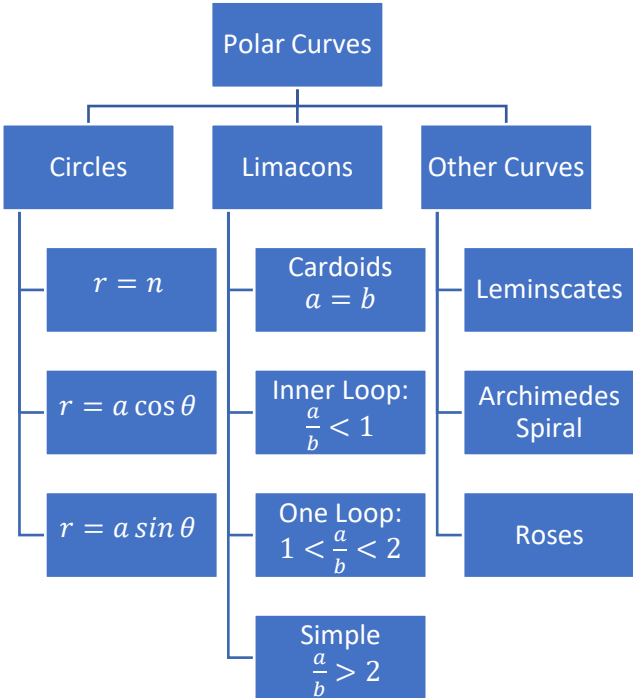
$0 \leq \theta \leq 2\pi$

$r = \sin n\theta, n \text{ is odd} \Rightarrow n \text{ petals}$
 $0 \leq \theta \leq \pi$

Example 2.93



The table shows a summary of some important types of polar curves.



F. Symmetry

2.94: Graphing Roses

Symmetry Tests for Polar Graphs in the Cartesian xy -Plane

1. *Symmetry about the x -axis:* If the point (r, θ) lies on the graph, then the point $(r, -\theta)$ or $(-r, \pi - \theta)$ lies on the graph (Figure 11.27a).
2. *Symmetry about the y -axis:* If the point (r, θ) lies on the graph, then the point $(r, \pi - \theta)$ or $(-r, -\theta)$ lies on the graph (Figure 11.27b).
3. *Symmetry about the origin:* If the point (r, θ) lies on the graph, then the point $(-r, \theta)$ or $(r, \theta + \pi)$ lies on the graph (Figure 11.27c).

2.4 Trigonometric Series

A. Trigonometric Series

2.95: Arithmetic Series

$$S = \alpha + (\alpha + \beta) + (\alpha + 2\beta) + [\alpha + (n - 1)\beta]$$

Where

$$\begin{aligned} \text{First Term} &= \alpha \\ \text{Common difference} &= \beta \end{aligned}$$

If we take an arithmetic sequence, and apply a trigonometric function to each term of the series, we will get a trigonometric series.

2.96: Sum of Sine Series

The sum of a trigonometric series with n terms:

$$S = \sin \alpha + \sin(\alpha + \beta) + \cdots + \sin(\alpha + (n - 1)\beta) = \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \sin\left(\alpha + \frac{(n - 1)\beta}{2}\right)$$

Step I: Expand

Begin with the given expression. Let:

$$S = \sin \alpha + \sin(\alpha + \beta) + \cdots + \sin(\alpha + (n - 1)\beta)$$

Multiply both sides by $2 \cdot \sin\left(\frac{\beta}{2}\right)$. The reason for this manipulation will become clear in the next step:

$$S \times 2 \cdot \sin\left(\frac{\beta}{2}\right) = \underbrace{2 \sin \alpha \sin\left(\frac{\beta}{2}\right)}_{1st \text{ Term}} + \underbrace{2 \sin(\alpha + \beta) \sin\left(\frac{\beta}{2}\right)}_{2nd \text{ Term}} + \cdots + \underbrace{2 \sin(\alpha + (n - 1)\beta) \sin\left(\frac{\beta}{2}\right)}_{N^{th} \text{ Term}}$$

Step II: Telescope

Use the product to sum formula ($2 \sin a \sin b = \cos(a - b) - \cos(a + b)$) to expand each term:

$$1st \text{ Term: } 2 \sin \alpha \sin\left(\frac{\beta}{2}\right) = \cos\left(\alpha - \frac{\beta}{2}\right) - \cos\left(\alpha + \frac{\beta}{2}\right)$$

$$2nd \text{ Term: } 2 \sin(\alpha + \beta) \sin\left(\frac{\beta}{2}\right) = \cos\left(\alpha + \frac{\beta}{2}\right) - \cos\left(\alpha + \frac{3\beta}{2}\right)$$

⋮

$$N^{th} \text{ Term: } 2 \sin(\alpha + (n-1)\beta) \sin\left(\frac{\beta}{2}\right) = \cos\left(\alpha + \frac{(2n-3)\beta}{2}\right) - \cos\left(\alpha + \frac{(2n-1)\beta}{2}\right)$$

The second term of each expansion cancels with the first term of the next expansion:

$$S \times 2 \cdot \sin\left(\frac{\beta}{2}\right) = \left[\cos\left(\alpha - \frac{\beta}{2}\right)\right] - \cos\left(\alpha + \frac{(2n-1)\beta}{2}\right)$$

Step III: Convert back to sin form

$$2 \sin a \sin b = \cos(a - b) - \cos(a + b)$$

Apply the reverse of the product to sum formula to the above expression. Find a and b such that:

$$\underbrace{a - b = \alpha - \frac{\beta}{2}}_{\text{Equation I}}, \quad \underbrace{a + b = \alpha + \frac{(2n-1)\beta}{2}}_{\text{Equation II}}$$

This is a system of equations in two variables, which we can solve using standard techniques., Add the two equations to eliminate b from the system:

$$2a = \alpha - \frac{\beta}{2} + \alpha + \frac{(2n-1)\beta}{2} = 2\alpha + (n-1)\beta$$

Solve for b by substituting $a = \alpha + \frac{(n-1)\beta}{2}$ in the first equation:

$$b = a - \left(\alpha - \frac{\beta}{2}\right) = \alpha + \frac{(n-1)\beta}{2} - \left(\alpha - \frac{\beta}{2}\right) = \frac{(n-1)\beta}{2} + \frac{\beta}{2} = \frac{n\beta}{2}$$

Step III: Final Answer

Substitute the values of a and b in the equality above:

$$S \times 2 \cdot \sin\left(\frac{\beta}{2}\right) = 2 \sin\left(\frac{n\beta}{2}\right) \sin\left(\alpha + \frac{(n-1)\beta}{2}\right)$$

Isolate and solve for S :

$$S = \frac{\sin\left(\frac{n\beta}{2}\right)}{\sin\left(\frac{\beta}{2}\right)} \sin\left(\alpha + \frac{(n-1)\beta}{2}\right)$$

3. 3D TOPICS

3.1 Parametric Equations

A. Cylinder

Example 3.1

Parametrize the cylinder with center at the origin, and radius 3.

The base of the cylinder is a circle with radius 3. Hence:

$$x^2 + y^2 = 9$$

Substitute $x = r \cos t, y = r \sin t$

$$\begin{aligned}(r \cos t)^2 + (r \sin t)^2 &= 9 \\ r^2(\cos^2 t + \sin^2 t) &= 9 \\ r^2 &= 9 \\ r &= \pm 3\end{aligned}$$

Substitute $r = 3$:

$$x = 3 \cos t, \quad y = 3 \sin t$$

There is no restriction on z ¹. Hence, the final parameterization is simply:

$$(x, y) = (3 \cos t, 3 \sin t)$$

3.2: Parametric Equation of a Circle

The equation of a circle with center $(0,0)$ and radius r is:

$$x = r \cos t, \quad y = r \sin t$$

Example 3.3

Parametrize the cylinder with center at the origin, radius 5, height 7, and whose maximum z value is 4.

The parametrization for the circular base is:

$$(x, y) = (5 \cos t, 5 \sin t)$$

Given the height of the cylinder is 7, and the maximum z value is 4, the minimum z value is

$$4 - 7 = -3$$

Hence, we must have:

$$-3 \leq z \leq 4$$

Let $z = t$:

$$-3 \leq t \leq 4$$

Hence, the final parametrization is:

$$(x, y, z) = (5 \cos t, 5 \sin t, t), \quad -3 \leq t \leq 4$$

Example 3.4

Suppose in the above example, t is a time parameter, and you need $t \geq 0$. Hence, find the parametrization.

$$-3 \leq z \leq 4$$

Add 3 throughout:

$$0 \leq z + 3 \leq 7$$

Substitute $z + 3 = t \Rightarrow z = t - 3$:

$$0 \leq t \leq 7$$

Hence, the final parametrization is:

$$(x, y, z) = (5 \cos t, 5 \sin t, t - 3), \quad 0 \leq t \leq 7$$

¹ Just like in 2D geometry, the line $x = 2$ is a vertical line that takes all values of y .

Example 3.5

Suppose in the above example, you further need the parametrization to go exactly once around the circle. Hence, find the parametrization.

To make the parameterization go once exactly once around the circle, we need:

$$0 \leq t \leq 2\pi, \quad 0 \leq z + 3 \leq 7$$

Consider a line in the tz plane with $(t_1, z_1) = (0, -3)$, $(t_2, z_2) = (2\pi, 4)$

$$\text{Slope} = \frac{z_2 - z_1}{t_2 - t_1} = \frac{4 - (-3)}{2\pi - 0} = \frac{7}{2\pi}$$

Use point-slope form $z - z_1 = m(t - t_1)$:

$$z + 3 = \frac{7}{2\pi}(t - 0) \Rightarrow z = \frac{7}{2\pi}t - 3$$

3.6: Parametrization Options

The more conditions you impose (and the more complex those conditions are), the more complicated the parametrization will be.

If your parameterization does not have any restrictions, you can simply use

$$z = t$$

As seen above, we had:

$$\begin{aligned} z &= t, & -3 \leq t \leq 4 \\ z &= t - 3, & 0 \leq t \leq 7 \\ z &= \frac{7}{2\pi}t - 3, & 0 \leq t \leq 2\pi \end{aligned}$$

Example 3.7

Consider the parametrization for the cylinder with center at the origin, radius 5, height 2, and whose maximum z value is 1:

$$(x, y, z) = (5 \cos t, 5 \sin t, t), \quad -1 \leq t \leq 1$$

Is it correct?

For the circle we to be complete, we need:

$$0 \leq t \leq 2\pi$$

Hence, the circle will not be complete in the above, which makes the parametrization wrong.

3.2 Further Topics**A. Hyperbolic Functions**

These are covered in the Note on Logarithms

8 Examples